

**MOLECULAR VARIABILITY IN ISOLATES OF
Rhizoctonia bataticola CAUSING
ROOT ROT IN CHICKPEA**

THESIS

**Submitted to
Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola
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**BY
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DECLARATION OF STUDENT

I hereby declare that the experimental work and its interpretation in the thesis entitled "**MOLECULAR VARIABILITY IN ISOLATES OF *Rhizoctonia bataticola* CAUSING ROOT ROT IN CHICKPEA**" or part thereof has neither been submitted for any other degree or diploma of any University, nor the data have been derived from any thesis / publication of any University or scientific organization. The source of materials used and all assistance received during the course of investigation have been duly acknowledged.

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Date : / /2017

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CERTIFICATE

This is to certify that thesis entitled "**MOLECULAR VARIABILITY IN ISOLATES OF *Rhizoctonia bataticola* CAUSING ROOT ROT IN CHICKPEA**" submitted in partial fulfilment of the requirement for the degree of "**Master of Science in Agriculture (Plant Pathology)**" of Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola is a record of bonafide research work carried out by **Gadekar Archana Arun** under my guidance and supervision.

The subject of the thesis has been approved by the Student's Advisory Committee.

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(D)**Abbreviations**

Abbreviations	Full form
%	: Per cent
/	: Per
@	: At the rate of
°C	: Degree Celsius
µg	: Micro grams
µl	: Micro litre (s)
AFLP	: Amplified Fragment Length Polymorphism
Cm	: Centimetre(s)
CTAB	: Cetyl Trimethyl Ammonium Bromide
dd H ₂ O	: Double distilled water
Deptt.	: Department
DNA	: Deoxyribose Nucleic Acid
dNTPs	: Deoxyribo nucleoside triphosphate
e.g.	: Exempli gratia (For example)
EDTA	: Ethylene Di-amine Tetra Acetic Acid
<i>et al.</i>	: Et alia (and associates)
etc.	: Et cetera
Fig.	: Figure
G	: Grams
i.e.	: That is
ISSR	: Inter Simple Sequence Repeat
SSR	: Simple Sequence Repeat
Kbp	: Kilo base pairs
M	: Molar
Max.	: Maximum
Mg	: Milli gram(s)
Min	: Minutes
ml	: Milli litre(s)

mM	:	Milli molar
Mm	:	Millimetre
Ng	:	Nano gram(s)
Nm	:	Nano meter
No.	:	Number
PCR	:	Polymerase Chain Reaction
PDA	:	Potato Dextrose Agar
PDB	:	Potato Dextrose Broth
ITS	:	Internal Transcript Spacer
RFLP	:	Restriction Fragment Length Polymorphism
rpm	:	Revolution per minute
<i>Taq</i> polymerase	:	<i>Thermus aquaticus</i> DNA polymerase Enzyme
TBE buffer	:	Tris hydroxymethyl amino methane and boric acid ethylene diamine tetra acetic acid buffer.
TE buffer	:	Tris hydroxymethyl amino methane ethylene diamine tetra acetic acid buffer
Tris	:	Tris Hydroxymethyl aminomethane
U	:	Unit
UPGMA	:	Unweighted pair group method for arithmetic mean average
UV	:	Ultra violet
<i>viz.</i>	:	Videlicet (namely)

(F) Thesis Abstract

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ABSTRACT

Chickpea (*Cicer arietinum*) is the world's third most important pulse crop. The major constraints limiting chickpea production is difficulty in managing diseases caused by soil borne pathogens viz., dry root rot (*Rhizoctonia bataticola*), *Fusarium* wilt (*Fusarium oxysporum* f.sp. *ciceri*). The losses in chickpea due to *Rhizoctonia bataticola* may be as high as 60 per cent.

The investigation was carried out in the laboratory of the Department of Plant Pathology and Biotechnology Centre, Dr. PDKV, Akola. The eight isolates of *Rhizoctonia bataticola* were collected from different districts of Maharashtra viz., Akola, Nagpur, Amravati, Satara (Karanje), Satara (Umbraj), Sangli, Sangli (Kurundwad) and Kolhapur (Kini).

The pathogenicity test of isolates of *Rhizoctonia bataticola* isolated from the different parts of the Maharashtra region was tested by using susceptible cultivar JG-62. On the basis of per cent mortality out of these eight isolates 2 (Rb-3, Rb-7) were strongly pathogenic, 6 (Rb-1, Rb-2, Rb-4, Rb-5, Rb-6, Rb-8) were moderately pathogenic.

The genetic diversity was studied amongst the eight isolates of *Rhizoctonia bataticola* using fifteen RAPD primers. Among them 10 primers produced 101 scorable bands. Out of 101 bands, 85 bands were polymorphic and average level of polymorphism was 84.16%. The primer OPB-14 amplified maximum 14 bands within the size 1951.44bp to 336bp, while primer OPB-11 and OPF-4 amplified minimum of 6 bands within the size 2159.78bp to 623bp and 950bp to 262.17bp, respectively.

It shows that Rb-4 (Umbraj) have higher value of similarity coefficient with Rb-5 (Karanje) whereas Rb-1 (Akola) have lower value of similarity coefficient with Rb-4 (Umbraj).

In the Dendrogram, the isolates of *Rhizoctonia bataticola* were grouped into two major clusters. First group is named as cluster-A, includes isolates belonging to Akola, Kurundwad, Nagpur and Kini. Second group is named as cluster-B, includes Amravati, Umbraj, Karanje and Sangli.

The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 34.92 to 72.55 per cent similarity coefficient for RAPD primers.

Genetic diversity was studied among the eight isolates using 6 SSR primers of MB series. Six primers produced 54 bands. Out of 54 bands, 43 bands were polymorphic and average level of polymorphism was 79.62%. The primer MB-14 amplified maximum of 14 bands within the size 130.40bp to 575.82bp, while primer MB-9 amplified minimum of 4 bands

within the size 100.16bp to 410.10bp, respectively. The 4 SSR primers MB-2 (300.45bp, 490.87bp, 675.39bp), MB-5 (220.54bp, 461.82bp), MB-14 (130.40bp, 160.57bp, 210.15bp, 310.45bp, 390.88bp), MB-17 (150.07bp) showed 11 monomorphic bands. Primers MB-9 and MB-18 showed 100% polymorphic bands.

In the Dendrogram, Rb-1 (Akola) was found to have higher value of similarity coefficient (0.7647) whereas Rb-3 (Amravati) was found to have lower value of similarity coefficient (0.3529). The isolates of *Rhizoctonia bataticola* were grouped into two major clusters. First group is named as cluster-A, includes isolates belonging to Akola, Kurundwad, Nagpur, Kini, Karanje and Sangli. Second group is named as cluster-B, includes Amravati and Umbraj.

It shows that Rb-1 (Akola) have higher value of similarity coefficient with Rb-7 (Kurundwad) whereas Rb-3 (Amravati) have lower value of similarity coefficient with Rb-8 (Kini).

The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 35.29-76.47 per cent similarity coefficient for SSR primers.

CHAPTER I

INTRODUCTION

1.1 Background Information

Chickpea (*Cicer arietinum* L.) is a major pulse crop of India. The centre of origin is southeastern Turkey. The earliest chickpeas were found 7500 years ago in Turkey and were known by the name “falcon face” in ancient Egypt. The genus *Cicer* belongs to the family Leguminosae and sub-family Papilionoideae. Most of the *Cicer* species are diploids with $2n=16$ chromosomes. There are two distinct types of chickpea, “kabuli” (Macrosperma) and “desi” (Microsperma) differing in their geographic distribution and different plant types. Kabuli types are usually taller and have large big or cream-coloured seed and “ram's head” seeds with white flowers and contain no anthocyanin. Desi types are generally shorter possessing small leaflets, pods, seeds. The seed colour varies from cream, black brown, yellow to green and predominantly with purplish flower and contain anthocyanin.

Chickpea commonly known as Gram or Bengal gram is the most important pulse crop used for human consumption as dal, vegetables etc. and straw as animal fodder. Chickpea is considered to have medicinal effects and it is used for blood purification. Chickpea is a highly nutritious grain legume crop. It is an important source of energy, protein, minerals, vitamins, fibers and other potentially health beneficial phyto-chemicals. Chickpea is valued for its nutritive seeds with 25.3-28.9 per cent protein content, 38-59 per cent carbohydrate, 3 per cent fiber, 4.8-5.5 per cent oil, 3 per cent ash, 0.2 per cent calcium and 0.3 per cent phosphorus (Hulse,1991; Huisman and Vander Poel, 1994).

The production of chickpea is concentrated in central and southern parts of India. Production trends during the past 15 years indicated that except for a few years in the early 2000s, the output of chickpea fluctuated at around 6 million tonnes for over a decade till 2007-

08. However, production has steadily increased thereafter and largely in tune with the trends in yield levels and reached about 8.8 million tonnes in 2012-13. This increase in production is possible due to development of high yielding disease resistant varieties, production and protection technologies and better rainfall in chickpea growing areas. Incidentally, the period the steep rise in production is in perfect line with the rise in minimum support prices (MSP) announced by the Govt. of India along with a significant expansion in the area under cultivation.

During 2012-13, highest production of chickpea (3551 thousand tonnes) was recorded in Madhya Pradesh with productivity of 435 kg/ha. In Rajasthan, production of 1265 thousand tones was achieved with productivity of 1010 kg/ha. In Maharashtra, production of 1058 thousand tones was recorded with productivity of 844 kg/ha. Highest productivity of 1430 kg/ha was recorded in Bihar state. In Andhra Pradesh, 759 thousand tonnes of chickpea was produced with productivity of 1115 kg/ha. (AICRP Chickpea). During 2015-16, production of chickpea (7060 thousand tonnes) was recorded in India.

Rhizoctonia bataticola (Taub.) Butlar (1925) (= *Sclerotium bataticola*) Taub. (1913) (Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid. (1947) is a diverse omnipresent ubiquitous soil-borne fungal pathogen, infecting more than 500 plant species. The pathogen causes different types of diseases viz., seed rot, seedling blight, root rot, charcoal rot, wilt, stalk rot, stem blight, fruit rot, seedling decay and leaf blight in crop plants (Dhingra and Sinclair, 1978). *Rhizoctonia* is a genus of anamorphic fungi in order Cantharellales and family Ceratobasidiaceae. *Rhizoctonia* species do not produce spores, but are composed of hyphae and sclerotia (hyphal propagules), asexual stage of fungi. *Rhizoctonia* species are saprophytic, but some act as facultative plant pathogens causing commercially important crop diseases. The genus *Rhizoctonia* (means “root killer”) was described in 1815 by the French mycologist Augustin Pyramus de candolle for plant pathogenic fungi that produce both hyphae and sclerotia.

The dry root rot in chickpea was first reported from India by Mitra (1931). *Rhizoctonia bataticola* causes up to 60% yield loss. High day temperature above 30°C and dry soil conditions at flowering and podding stages rapidly increase the severity of the dry root rot disease on chickpea. Singh and Sharma (2002) observed that deficit soil moisture favours the severe disease development on pulse crops. Under hot and dry environmental conditions, many economically important crops are predisposed to the infection and colonisation of *Rhizoctonia bataticola* and caused drastic yield losses on chickpea.

The dry root rot symptoms are most commonly observed in chickpea during post-flowering stage which include drooping and chlorosis of petioles and leaflets, initially confined to top leaves of the plant (Mamta Sharma and Pande, 2013). Leaves and stems of affected plants are usually straw coloured and in some cases, the lower leaves and stems are brown. The tap root turns black with signs of rotting and is devoid of most of the lateral and finer roots. The dead roots are quite brittle and show shredding of the bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil when plants are uprooted. Dark minute sclerotial bodies can be seen on the roots exposed and inner side of the bark or when split open at the collar region vertically.

1.2 Importance of study

Among the various existing diverse climatic situations and environmental variations that are responsible for low yields, resurgence of diseases of crop plant are one of the most important factors. According to survey conducted in countries of the world in the year 1995, the no. of pathogens causing various diseases in chickpea had been reported to be 172, which include 67 fungi, 3 bacteria, 22 viruses and mycoplasma and 80 nematodes (Nene *et al.*, 1996). In India, this crop suffers from serious diseases like Wilt (*Fusarium oxysporum* f. sp. *ciceri*), Dry root rot (*Rhizoctonia bataticola*), Ascochyta blight (*Ascochyta rabiei*), Gray mold (*Botrytis cinerea*), stunt virus etc.

Due to infection of dry root rot disease a loss to the extent of 10-20 per cent was reported by Dhar and Chaudhary, (2001). *Rhizoctonia bataticola* causes up to 60% yield loss (Subramanian Sundravada *et al.*, 2012).

The number of simple techniques were known that could be used to rapidly characterize *Rhizoctonia bataticola* population in a particular area. However, evaluating genetic diversity information for *Rhizoctonia* management in field requires powerful discriminating, selective and reliable criteria for genotyping the isolates (Sharma *et al.*, 2004). In order to measure genetic variability more precisely, molecular markers provide an unbiased estimate to total genomic variation and have the potential to minimize errors due to sampling variance (Spooner *et al.*, 1996). Furthermore, determination of fungal genetic diversity based on molecular markers is reliable and independent technique to assess variability in the population of the pathogen.

Characterization of diverse populations of *Rhizoctonia bataticola* could improve the knowledge of population biology and its interaction to contribute to the design of optimal breeding methods to produce chickpea cultivars with stable and durable resistance to dry root rot. In addition to morphological and physiological variation, genetic variability in plant pathogens may be of great importance to minimize disease incidence. Breeding for disease resistant plant with durable resistant requires that the plants should be resistant to all pathotypes of the pathogen. Highly variable plant pathogens may be difficult to combat the development of resistant plant genotypes. The DNA based techniques called Random Amplified Polymorphic DNA (RAPD) (Williams *et al.*, 1990) constitutes one of the methods for discrimination among strains of species. It compares favourably with strains of species and is often deemed superior in resolution or efficacy to traditional traits. This will be a valuable tool for epidemiological studies of these fungi and may be important for the development of selective pathogen management strategies.

1.3 Objective of the study

Dry root rot incited by *Rhizoctonia bataticola* is emerging as a serious constraint to chickpea production as the disease turns epidemic. An intensive rapid roving field survey was conducted to assess the distribution and incidence of *Rhizoctonia* root rot and to collect the infected plant samples in major chickpea growing district viz., Akola, Nagpur, Amravati, Satara, Sangli and Kolhapur during *rabi* season from soybean gram cropping system. The present investigation was carried out to assess the extent of genetic variability among *Rhizoctonia bataticola* isolates, collected from different area with following objective:

1. To analyse the molecular variability among isolates of *Rhizoctonia bataticola* causing chickpea root rot.

1.4 Hypothesis

Increased attention is being paid by the scientists to develop dry root rot resistant varieties for large scale cultivation of chickpea and appropriate management of the disease. Therefore accurate and rapid identification of pathogen is necessary. Identification of pathogenic races would be done by use of different reaction to selected host genotypes. To reduce the effects caused by the variability in the pathogen it is essential to know the genetic nature and pathogenic types of isolates prevailing in chickpea growing areas of India. Hence the present investigation will help to understand the variability of pathogen which in turn will be useful for its exploitation in dry root rot resistance breeding and further used for development of area specific resistant varieties of chickpea. The result of present investigation demonstrate that applicability of RAPDs of genomic DNA in analyzing genetic variation, which are particularly useful when combined with *in vitro* and *in planta* studies. Also, this study provides the detailed description of the diversity existing in the *Rhizoctonia bataticola* isolates in different geographical locations in Maharashtra that will be useful for breeders to identify the location-specific *Rhizoctonia bataticola* resistant cultivars and facilitate the selection of germplasm to incorporate in advance breeding strategies for more or less tolerance to dry root rot in a given geographic region.

1.5 Scope and Limitation

Looking into the nature of pathogens involved, there is no alternative but to go for developing resistant varieties of chickpea rather than the excessive use of fungicides for management of this disease. As, field application of excessive fungicides, often is the case, pose increased risk of evolution of resistant strains, ground water contamination, human health, in addition to affecting beneficial soil micro flora and long term soil fertility, screening for breeding of resistant cultivars should be strongly advocated.

In the recent year, it was observed that due to presence of pathogenic variability in the pathogen at the one location results in the breakdown of resistance in several chickpea genotypes. There are limited root rot resistant genotypes having good agronomical characters. In such situation, the resistant genotypes need to be used precisely in breeding programme. The past report suggests that host plant resistance does not persist for long term as variety presumed resistance to root rot failed in near future either as a result of genetic breakdown or change in the virulence of the pathogen. Hence, there is great scope to establish relation with pathogenic variability to find out the possible level of genetic diversity through molecular markers which could be applicable in introduction of resistance breeding programme.

Hence, it would be necessary to ascertain the genetic and pathogenic variability amongst the pathogen. In view of the above, an investigation was carried out on “Molecular Variability in isolates of *Rhizoctonia bataticola* causing root rot in chickpea” during 2015-2016 at Department of Plant Pathology, Post Graduate Institute, Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

CHAPTER II

REVIEW OF LITERATURE

Literature pertaining to present investigation reviewed critically and arranged under following major heads.

2.1 *Rhizoctonia bataticola*

Rhizoctonia is a genus of anamorphic fungi in the order Cantharellales and family Ceratobasidiaceae. The genus *Rhizoctonia* (means “root killer”) was described in 1815 by the French mycologist Augustin Pyramus de Candolle for plant pathogenic fungi that produce both hyphae and sclerotia. *Rhizoctonia* species are saprophytic, but some act as facultative plant pathogens causing commercially important crop diseases. The dry root rot in chickpea was first reported from India by Mitra (1931).

Nene *et al.* (1979) reported that dry root rot of chickpea caused by *Rhizoctonia bataticola* and sclerotia can be seen under the root bark which peels off easily.

Nene *et al.* (1981) observed that dry root rot has been reported from Australia, Ethiopia, India, Iran and USA. The disease is serious in areas where the ambient day temperature are around 30⁰ C at the seedling or at the flowering stage. In India this disease occurs mainly in the central and southern states where a lot of mortality at the flowering and podding stages is due to this disease.

Trapero-Casas and Jimenez-Diaz (1985) observed that dry collar and root rot occurred most frequently at the seed development stage in chickpea. Stems and leaves dried up and become straw coloured.

2.2 Occurrence and losses caused by *Rhizoctonia bataticola*

Taya *et al.* (1988) stated that the incidence of *Rhizoctonia bataticola* (*Macrophomina phaseolina*) on *Cicer arietinum* was higher in sandy than clay soil. Low soil moisture was conducive to disease development.

Pandey and Singh (1990) conducted the survey in Allahabad region of Uttar Pradesh and reported that the chickpea was infected by *Rhizoctonia bataticola* in 83% of villages with less than 25% disease intensity.

Than *et al.* (1991) stated that *Rhizoctonia bataticola* isolates of chickpea were collected from several location of Myanmar observed that the possible occurrence of more than one type of isolate in same field then the existence of closely related isolates from different places.

Prajapati *et al.* (2003) observed that the incidence of dry root rot in chickpea caused by *Rhizoctonia bataticola* was studied in 20 locations in Uttar Pradesh and Uttaranchal, India during February to March of 1998-2000 and reported that disease was widespread in Uttar Pradesh. It was severe in bold seeded cultivars.

Shubha and Gurha (2006) conducted the survey in different blocks of Bhundelkhand region of Uttar Pradesh found infection of *Rhizoctonia bataticola* on chickpea ranged from 5-22 %.

Gurha and Trivedi (2008) Observed that chickpea fields in Shimoga, Raichur and Bangalore districts in Karnataka were surveyed for the prevalence of dry root rot. *Rhizoctonia bataticola* was the predominant pathogen infecting 60 to 70% of the plants in the fields of Gulbarga, Manchalapur, Ekhaspur, Raichur and Tengri.

Manjunatha *et al.* (2011) conducted intensive roving survey during November to December, 2008 in the farmers' fields at Raichur, Gulbarga and Bidar districts in Karnataka. The maximum dry root rot incidence in chickpea was noticed in Gulbarga district (9.8%) followed by

Raichur (7.6%) and the least (6.18%) in Bidar. The overall incidence of dry root rot ranged from 1-19% across the districts. This was mainly attributed due to variation in soil type (black and red soil) and cultivars grown (Annegiri and JG 11) under rainfed condition.

Khan *et al.* (2012) conducted survey for the occurrence and severity of chickpea dry root rot was made during crop season 2010-2011 in Jammu and Kashmir. The highest incidence was up to 40% in Shangus and lowest up to 4.11% in Naina of Jammu and Kashmir. The incidence of dry root rot in chickpea caused by *Rhizoctonia bataticola* was observed in late October to mid November. The intensity of the disease was high in the month of February and March during late flowering and podding stage.

Ghosh *et al.* (2013) conducted a survey in 2010-2011 *rabi* cropping season to obtain information on the distribution and incidence of chickpea diseases with respect to soil type and cultivar used. They found dry root rot and collar rot diseases at all of the sites with incidence ranged from 8.9% - 10.3% and 7.1% - 10.5%, respectively irrespective of cultivar type and locations.

2.3 Survey for incidence of *Rhizoctonia* root rot

Rhizoctonia bataticola (Taub.) Butler is one of the most damaging pathogens in arid region and has a wide host range (Waller, 1976). In India and particularly from Andhra Pradesh, Bihar, Gujarat, Karnataka, Maharashtra, Madhya Pradesh, Rajasthan, Tamil Nadu, Uttar Pradesh and West Bengal (Uppal, 1931; Likhite and Kulkarni, 1934; Deshpande *et al.*, 1969 and Philip *et al.* 1969).

Gupta *et al.* (1983) reported that the incidence of dry root rot of chickpea ranging from 3.2 to 20.6 % in 30 villages of northern Madhya Pradesh.

Saifulla *et al.* (2013) carried out survey to know the severity of various diseases on chickpea during *rabi* season of 2010-2011 and recorded maximum dry root rot incidence in Bangalore urban areas

(26.40%) followed by Chikmagalur (6.56%), Hassan (5.48%), Chamrajnagar district (5.41%) and Davanagere (4.58%). The least disease incidence was recorded in Chitradurga (0.40%) whereas Tumkur district was free from disease.

Nyandoro *et al.* (2014) conducted a survey in August 2014 in southern Alberta of Canada to determine the incidence and severity of root rot in soybean fields and observed that root rot occurred in all locations with incidence averaging 34% and ranging from 18% to 73% and severity averaging 0.9 and ranging from 0.3 to 1.2 on a scale of 0-4.

2.4 Symptoms and pathogenicity

Dastur (1935) described that only mature plants were affected, showing bronzing of the leaves on one or more of the lower branches. The colour of leaves later changed to yellow and then brown. The affected branches and leaf stalks of diseased plants were stiff and turned upwards and leaflets stand more or less vertically and shed prematurely. The terminal part of the tap root was black or brown and shriveled.

Kotasthane *et al.* (1980) reported that drying of chickpea plants was observed due to several soil borne pathogens including *Fusarium oxysporium* f. sp. *ciceri*, *Rhizoctonia bataticola*, *Sclerotium rolfsii* and *Opercullella padwickii*.

Singh *et al.* (1990) reported that roots inoculated with *Rhizoctonia bataticola* (*Macrophomina phaseolina*) caused disintegration of cortical tissues with mycelial and sclerotial bodies plugging the xylem vessels.

Nene *et al.* (1991) described that the dried plants due to disease generally appeared during flowering and podding time. These were scattered in the field. The seedling can also get infected. Drooping of petioles and leaflets was confined to those at top of the plant. Leaves and stem of affected plant usually straw coloured but in some cases, the lower

portion of the stem was brown. When the plant was uprooted, lower portion of the tap root remains in the soil. It was devoid of lateral and finer roots. Dark minute sclerotial bodies can be seen on the roots exposed inside the wood.

Jayanti and Bhatt (1993) observed that sown pre-germinated seed of 21 chickpea varieties in soil inoculated with *Rhizoctonia bataticola* which caused seed rot within 24 hr. in NEC874 and EG234. Bold 2375, BG209, JG62, JG315, ICC 3375 and JG1133 developed necrotic lesion in 3-5 cells deep at hypocotyl region within 7 days. RSG-44, AGC677, NEC41, GL269, JG74, ICC8983 and ICC 5003 developed only superficial necrosis along the hypocotyl region. BMG 416, BG 416, ICC 1376 and ICC 1376 and ICC 113314 were resistant. Resistant cultivars had a greater number of lateral roots during early growth phases.

Pande *et al.* (2004) observed that dry root rot of chickpea caused by *Rhizoctonia bataticola* generally appears during late flowering and podding stages and the infected plants appear completely dried. The root system of diseased plant shows extensive rotting with most of the lateral roots are brittle and minute sclerotial bodies appear in the pith cavity and on the outer surface of the tap root.

Meena *et al.* (2006) reported that charcoal rot of maize caused by *Macrophomina phaseolina* is favoured by high soil temperature ranging from 30 to 42⁰ C and low soil moisture. The pathogen over-winters as sclerotia in soil and upon germination might penetrate roots and lower stems during growing season. Also a characteristic sign is the presence of numerous, minute and black sclerotia, particularly on the vascular bundles and outside the rind of the bark.

Monga *et al.* (2007) tested pathogenicity of *Rhizoctonia bataticola* isolates on *Gosypium arborium*, desi cotton variety, RG-8. High mortality was recorded in isolate RJ-28 (83.3%) followed by HR-3-6 (75.0%) whereas minimum mortality was observed in isolates HR-15, HR-16, HR-3-7, PB-2, HR-18, RJ-2, RJ-6, Rb-NCIPM and P-3-13.

Jayalakshmi *et al.* (2008) reported that pot culture experiment was conducted for screening 12 promising entries of chickpea (JG-11, GBS 964, GBS 963, GBC 2, GBM 6, GBM 2, GBM 10, BGD 103, GCP 107, GCP 101, A1, ICCV 10, L-550+1 susceptible check JG-62) against dry root rot during *rabi* season, 2004-05 and 2005-06. Out of 12 genotype tested against root rot disease, 4 resistant, 2 moderately susceptible, 3 susceptible and 3 highly susceptible to dry root rot disease were observed.

Aghakhani and Dubey (2009^a) conducted pathogenicity test of twenty-three isolates of *Rhizoctonia bataticola* (*Macrophomina phaseolina*) causing dry root rot of chickpea collected from 10 different major chickpea growing states of India. Highly susceptible cultivar Pusa 212 was susceptible and all the isolates were pathogenic under artificially inoculated conditions. Artificially inoculated plants expressed typical symptoms of dry root rot. The infected plants lost turgidity and showed chlorotic symptoms. The leaves and stems of infected plants turned straw coloured. Tap root became dark and rotting extended to entire root system. Under favorable conditions, minute sclerotia developed on root.

Mamta *et al.* (2012) performed pathogenicity test for 94 isolates of *Rhizoctonia bataticola* in laboratory on a susceptible cultivar BG 212 of chickpea by paper towel technique. All 94 isolates were highly pathogenic with disease severity ranging from 7 to 9 rating (1 to 9 scale), except two isolates [Jodhpur (RB 14) and Delhi isolate (RB 16) and Delhi isolate (RB 16)] with a disease severity of 4.

Khan *et al.* (2012) observed the incidence of dry root rot of chickpea caused by *Rhizoctonia bataticola* in late Oct. to mid Nov. with maximum incidence in the month of Feb. and March *i.e.* during late flowering and podding stage. The symptoms of the disease were yellowing of the leaves within a day or such leaves drop and plant showed completely dried symptoms within a week after the appearance of the first symptom. If the plant were pulled out from the soil and examined the basal stem and main root system of diseased plant showed extensive rooting

with most of the lateral roots decayed. The tissues were weakened and break off easily. In advance cases the sclerotial bodies may scattered in the pith cavity and on the outer surface of the tap root.

Mamta *et al.* (2015) observed the symptoms of dry root rot in chickpea during post-flowering stage which include drooping and chlorosis of petioles and leaflets, initially confined to top of the plant. Leaves and stems of affected plants are usually straw coloured and in some cases, the lower leaves and stems were brown. The dead roots were quite brittle and shredding of the bark.

2.5 Variability among isolates of *Rhizoctonia bataticola*

2.5.1 Cultural and morphological variability

Uppal *et al.* (1936) studied the temperature requirement for growth of *Rhizoctonia bataticola* isolates and reported that two isolates did not grow at 9°C and 11°C, respectively after 2 days, but slight growth of both isolates was observed after 5 days. They also recorded the optimum temperature between 30 and 35°C for the growth of isolates.

Hildebrand *et al.* (1945) observed the sclerotia were black, smooth varying from spherical to oblong to irregular shaped. The sclerotial shape of any isolate varied with substrate on which fungus was grown.

Ghosh and Sen (1973) observed that sclerotial diameter of *Rhizoctonia bataticola* in gram varied from 98.1-119.2 µm.

Arca and Yildizin (1989) reported that out of 362 isolates of *Rhizoctonia bataticola* of Aegean region, 333 exhibited variations in cultural characteristics, such as colony colour, mycelial growth, colony radius and the number of sclerotia. The group of isolates were established based on radius and the number of sclerotia.

Prameela Devi and Singh (1998) obtained thirty-six isolates of *Macrophomina phaseolina* to study cultural variation and categorized into three morphologically different groups. One representative isolate from each group viz., MP-1, MP-2 and MP-3 was selected for detailed

studies. MP-1 isolate had cottony mycelial growth with pale grey colonies. The colony colour of the isolate was brown in MP-2 and showed 5 μm appressed mycelial growth. The sclerotia were bigger in size and irregular in MP-2 (400 X 280 μm), oblong in MP-1 and round in isolate MP-3. Typical right-angled branching of mycelium found in isolates MP-2 and MP-3.

Monga *et al.* (2007) studied morphological characters of 23 *Rhizoctonia bataticola* isolates and exhibited the variation in colour ranged from creamish to brown mycelium. They also observed variation in formation of sclerotia. It varied from sclerotial initials in most cases and abundant minute to few large sclerotia in few isolates.

Aghakhani and Dubey (2009^c) reported that twenty-three isolates of *Rhizoctonia bataticola* (*Macrophomina phaseolina*) were highly variable in their morphological as well as cultural characters. The majority of the isolates (19) produced suppressed mycelium whereas 4 isolates produced aerial mycelium. The pigmentation of various colours as white to dull white, creamy, grey and black was prominent among the isolates. The isolates found variable in respect of their colony growth. After 48 hrs. of incubation colony growth ranged from 51.3 to 89.0 mm. The sclerotia formed in different isolates were dark brown to black and highly variable in size (40-600 μm).

Sundravadana *et al.* (2012) observed that 11 isolates of *Rhizoctonia bataticola* from different pulse crops viz., red gram, green gram, cowpea, soybean and black gram showed variability in sclerotial intensity, shape and intensity of pigment synthesis. All isolates were more aggressive on the original host from which it was isolated than on other host.

Mamta *et al.* (2012) collected *Rhizoctonia bataticola* populations from diverse agro-ecological zones. Different cultural and morphological parameters like colony colour, growth pattern, growth rate, mycelial characters, sclerotial initiation time, sclerotial intensity and

morphology of the sclerotia, were assayed. The colony colour of *Rhizoctonia bataticola* varied from light black to black and light grey to grey. The shape of colony varied from radial to irregular and production of aerial mycelium was too low in most of the isolates.

Khan *et al.* (2012) carried out studies to find out optimum temperature for growth and sporulation of the *Rhizoctonia bataticola*. Pathogen could grow over a wide range of temperature *i.e.* 10⁰C to 45⁰C and the optimum was at 30⁰C followed by 35⁰C, with no sclerotial production at any temperature. Further they stated that it could grow over a wide P^H range of 3.0 to 9.0 but the optimum P^H for its growth was 5.5. No sporulation of the fungus occurred at any P^H level tested.

2.5.2 Molecular variability by Random Amplified Polymorphic DNA (RAPD) and Simple Sequence Repeat (SSR)

It involves the use of single short random oligonucleotide sequence (called random primers) defined cyclic amplification of DNA, which exposes the polymorphism, distributed throughout the genome. The amplified fragments are called random amplified polymorphic DNA. (Williams *et al.*, 1990)

Vandemark *et al.* (2000) used a simplified protocol for detecting amplified restriction fragment length polymorphism (AFLP) to evaluate genetic diversity among isolates of *Macrophomina phaseolina* collected from different hosts and locations. Fifteen different selective primer pairs were evaluated. No significant differences were observed among primer pairs, grouped based on the AT % of the selective nucleotides, for several parameters that define primer utility. Relationship matrices generated with each group of AFLP primers were highly correlated ($r^2 > 0.92$). Isolates were very diverse and could not clearly be grouped based on the geographic locations from which they were obtained. Genetic relationships among isolates were very robust. Relationships were determined using both the genetic similarity method of Nei and Li and the 'genetic distance' method of Skroch. They concluded

that both methods were equally effective for determining intraspecific genetic relationships when the majority of markers were polymorphic.

Su *et al.* (2001) investigated variability among isolates of *Rhizoctonia bataticola* isolated from soybean, corn, sorghum and cotton root tissues by RFLP and RAPD analysis. No variations were observed among isolates in restriction pattern of DNA fragments amplified by PCR, covering internal transcribed spacer (ITS) region 5.8 rRNA and part of 25sRNA but variation were observed when ten random (RAPD) primers were used to amplify the total DNA of isolates.

Tarakanta *et al.* (2003) showed that single RAPD primer A13 could be used to identify and discriminate several isolates of *Macrophomina phaseolina* and *Fusarium* sp. obtained from 20 hosts including soybean, cotton, chickpea and safflower.

Almeida *et al.* (2003) studied Fifty-five isolates from soybean roots were collected in different regions and analyzed through RAPD for genetic diversity. The UPGMA cluster analysis for 74 loci scored permitted identification of three divergent groups with an average similarity of 99%, 92% and 88% respectively. Knowledge of this variation may be useful in screening soybean genotypes for resistance to charcoal rot.

Rajkumar (2004) observed that ten isolates of *Rhizoctonia bataticola* representing Dharwad and Bijapur region proved pathogenic to sorghum. Molecular profile using RAPD markers indicated the genetic differences among isolates were species specific finger print to *Rhizoctonia bataticola*. Isolates were grouped by adopting molecular and virulence data.

Jana *et al.* (2005) used single primer of simple sequence repeats (SSR) or microsatellite markers for the first time for the characterization of genetic variability of different populations of *Macrophomina phaseolina* obtained from soybean and cotton grown in India and USA. They calculated genetic similarity between the isolates and used cluster analysis to generate a dendrogram showing relationships

between isolates collected from two hosts. Forty isolates clustered into three major groups corresponding to their hosts and geographical region. The SSR fingerprints (0.25-3.5 kb) generated using DNA from different populations of *Macrophomina phaseolina* of two hosts indicated that these repeats were interspersed within the genome of the pathogen. The variability found within closely related isolates of *Macrophomina phaseolina* indicated that such microsatellites were useful in population studies and represented a step towards identification of potential isolate diagnostic markers specific to soybean and cotton.

Bogale *et al.* (2005) describe nine simple sequence repeat (SSR) markers developed for studying *Fusarium oxysporum*. Allelic diversity at the nine loci ranged from 0.003 to 0.895, with a total of 71 alleles among 64 isolates. These markers will facilitate studies on relationships amongst isolates of *Fusarium oxysporum*.

Reyes Franco *et al.* (2006) studied genetic characterizations of 96 isolates of *Macrophomina phaseolina* from Mexico and other countries to define pathogenic and genetic patterns of diversity on the basis of amplified fragment length polymorphism (AFLPs) by using four combinations of oligonucleotides. The analysis revealed 418 amplified products of which 7.2 % were monomorphic and 92.8 % were polymorphic. Cluster analysis of all the isolates showed the substantial genetic diversity in *Macrophomina*. Two major clusters of isolates were observed, the first (A) included 90 isolates from all countries with genetic dissimilarity from 3 % to 11 % while the second group (B) included six isolates from Argentina, Australia, USA (two isolates), Japan and Brazil.

Meena *et al.* (2006) analyzed seven isolates of *Macrophomina phaseolina* incitant of maize charcoal rot through RAPD marker for genetic diversity by using 100 primers (OPA-E) and reported that the UPGMA cluster analysis for 706 loci identified three main clusters. Similarity matrix and Jaccard's similarity coefficient between the isolates indicated the maximum genetic variation among isolates of Arabhavi and Coimbatore (70.8 %) followed by Ludhiana and Coimbatore (69.5 %). The

most closely related isolates were Hyderabad and Delhi with an affinity percentage of 65.5 followed by Udaipur and Bangalore isolates with a percentage of 62.9 % similarity.

Purkayastha *et al.* (2006) demonstrated that genetic variation among 59 isolates of *Macrophomina phaseolina* from cluster bean as well as other hosts using RAPD assay and PCR-RFLP of the ITS region, 5.8S gene and part of the 25S rDNA and reported that RAPD assay clearly distinguished the isolates on the basis of chlorate phenotype and host origin. Isolates from single host were generally similar to each other but differed distinctly from those from other hosts. Chlorate-sensitive isolates were distinct from chlorate-resistant isolates within a given host. Also, for the first time high degree of polymorphism in restriction patterns of the ITS region, including part of 25S rDNA had been reported in the charcoal rot fungus by them.

Rajkumar *et al.* (2007) studied genetic diversity among 10 isolates of *Macrophomina phaseolina*, a causal agent of charcoal rot disease in sorghum using PCR-RAPD markers. From a set of 40 random primers tested, amplicon profiles of 15 were reproducible. A total of 149 amplicon levels, with an average of 9.9 bands per primer, were available for analysis of which 148 were polymorphic (99.3%). UPGMA clustering of data indicated the isolates shared varied levels of genetic similarity within a range of 0.14 to 0.72 similarities coefficient index and grouping of isolates was not related to sampling location.

Monga *et al.* (2007) analyzed twenty five isolates of *Rhizoctonia bataticola* causal organism of cotton root rot using RAPD markers to study the relationship between molecular variability and pathogenicity. All the fifteen primers used generated scorable polymorphic bands for the isolates and classified the isolates into two broad groups. The group-1 consisted of isolate no. 16 having the genetic similarity 0.59 and found to be pathogenic (83.3%). Group-2 was further clustered into five sub-groups. The largest sub-groups were II b and II c which consisted of 9 isolates each with 0.79-1.0 genetic similarity coefficient. The other

sub-group IIa, IIc and IIe consisted of two isolates each having similarity coefficient ranged from 0.63-1.0. However, no strict correlation was observed between isolates grouped based on pathogenicity, morphological features and RAPD finger printing.

Aghakhani and Dubey (2009^b) determined genetic diversity among 27 isolates (23 from chickpea and 4 from other host crops) of *Rhizoctonia bataticola* representing 11 different states of India by random amplified polymorphic DNA (RAPD), internal transcribed spacer restriction fragment length polymorphism (ITS-RFLP) and ITS sequencing. The isolates showed variability in virulence test. They used unweighted paired group method with arithmetic average cluster analysis to group the isolates into distinct clusters. The clusters generated by RAPD grouped all the isolates into six categories at 40% genetic similarity. High level of diversity was observed among the isolates of different as well as same state. Some of the RAPD (OPN-4, OPN-12 and OPN-20) markers clearly distinguished majority of the isolates into the area specific groups. The ITS I, 5.8r DNA and ITS II regions of 11 isolates representing different RAPD groups were amplified with primers ITS-1 and ITS-4 and digested with seven restriction enzymes. The restriction enzymes Dra I, Mbo I, Rsa I and Alu I were found to be suitable for differentiating the isolates into five categories by showing isolate specific ITS-RFLP patterns.

Fiers *et al.* (2011) characterized the genetic diversity of 73 French strains of *Rhizoctonia solani* collected from the different potato growing areas in France by phylogenetic analyses of the internal transcribed spacer sequences of ribosomal RNA. The determination of AGs of *Rhizoctonia solani* based on the sequencing of these ITS region showed three different AGs among the collection (60 AG 3 PT, 8 AG 2-1 and 5 AG 5).

Mamta Sharma *et al.* (2012) collected 94 isolates from *Rhizoctonia bataticola* infected chickpea plants from different agro climatic regions of India to find out the diversity in *Rhizoctonia bataticola* populations. They analyzed the samples with amplified fragment length

polymorphism (AFLP). Five AFLP primer combinations provided a total of 121 fragments. All fragments were found polymorphic with an average polymorphic information content value of 0.213. The dendrogram based on AFLP analysis showed maximum number of *Rhizoctonia bataticola* isolates were very diverse and did not depend on geographical origin.

Pancheshwar *et al.* (2012) isolated the genomic DNA of 21 isolates of *Rhizoctonia bataticola* and subjected to amplification with RAPD markers for diversity analysis. Randomly selected 8 decamer primer amplified 64 RAPD marker loci. Out of these 64 bands, 29 bands (45.3%) were polymorphic and remaining 35 bands (54.6%) were monomorphic. The range of genetic diversity among *Rhizoctonia bataticola* isolates was 0.69 to 0.98. The cluster analysis showed 2 distinct groups among 21 isolates. They also suggested that any *Rhizoctonia bataticola* accessions under study did not made any group in respect to their geographical region.

Datta and Lal (2012) characterized genetic diversity in 14 isolates of wilt pathogen *Fusarium oxysporum* f. sp. *ciceri* collected from major pulse growing regions of India. A total of 24 alleles were produced by twelve simple sequence repeats primers with an average of two alleles per marker in *Fusarium oxysporum* f. sp. *ciceri* isolates. The maximum number of 4 alleles was obtained with primer SSR 12. SSR amplicon size ranged from 100 to 400 bp. The unweighted pair group method with arithmetic average (UPGMA) cluster analysis based on SSR profiles grouped the fourteen *Fusarium oxysporum* f. sp. *ciceri* isolates into four major clusters.

Bimal *et al.* (2013) investigated that 20 ISSR primers amplified 174 markers. Out of them 122 loci were polymorphic 70.11%, while 52 loci were monomorphic 29.89%. A dendrogram was generated by UPGMA clusters analysis based on Jaccard's similarity coefficient. Estimated genetic similarity among 18 soybean varieties using 20 ISSR primers, were 0.71-0.95.

Prasad *et al.* (2013) collected eight isolates of *Rhizoctonia solani* Kuhn, the incitant of wet root rot of chickpea from Haryana, Rajasthan and Delhi and assessed their genetic variability using RAPD markers. Isolates differed in their RAPD banding pattern. The similarity value of RAPD profiles ranged from 0.12 (Hanumangarh – Karnal) to 0.79 (Delhi – Gurgaon and Hisar – Karnal) with an average of 0.34 ± 0.1 among the isolates. They used ten random primers to fingerprint the individual isolates and using un-weighted pair group method with arithmetic average distinguished *Rhizoctonia solani* isolates into different fingerprint groups.

Mishra *et al.* (2015) did molecular characterization of *Rhizoctonia solani* (kuhn) isolates of three hosts, viz., rice, maize and green gram using Random Amplified Polymorphic DNA (RAPD) and reported that these markers were found suitable for resolving the genetic polymorphism hidden among 22 *Rhizoctonia solani* isolates. The phylogenetic analysis based on RAPD revealed three clusters comprising two major clusters of maize and rice and one minor cluster with four outliers of the isolates. The two green gram isolates, PRS21 and PRS22, were clustered with maize isolates.

Belkar and Gade (2016) isolated the genomic DNA of 20 isolates of *Rhizoctonia bataticola* and subjected to amplification with RAPD markers for diversity analysis. Randomly selected 20 decamer primer amplified 149 RAPD marker loci. Out of these 149 bands, 111 bands (74.49%) were polymorphic. Similarity coefficient values for 20 accessions of *Rhizoctonia bataticola* in RAPDs marker were 0.36 - 0.99.

2.5.3 Internal transcript spacer (ITS)

Chakrabarty *et al.* (2007) studied PCR amplification of *Rhizoctonia bataticola* with conserved primers ITS-1 and ITS-4 yielded ~600 base pair rDNA product which were cloned in plasmid pGEMT.

Aghakhani and Dubey (2009^b) studied six isolates of *Rhizoctonia bataticola* PCR amplification with primer set ITS-1 and ITS-4 yielded single DNA fragment present in all isolates with ~500-650 bp in size.

Ragini *et al.* (2013) analyzed 13 isolates of *Macrophomina phaseolina* were subjected to PCR amplification of ITS region and compassing 5.8s gene region using universal primers ITS-1 and ITS-4. All the representative isolates generated a single band of ~500 bp which included partial sequence of ITS-1, 5.8 gene, ITS-2 and partial sequence of 28s gene upon direct sequencing.

CHAPTER III

MATERIAL AND METHODS

During the course of present studies the details of materials used and methods implemented are presented in this chapter. All *In vitro* studies on root rot pathogen of chickpea and molecular analysis work were conducted in the Department of Plant Pathology, Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

3.1 Material required

3.1.1 Glasswares

Several types of glasswares viz., Petri plates, reagent bottles, conical flasks (viz., 100 ml, 250 ml, 500 ml, 1000 ml and 2000 ml), test tubes, micropipettes, beakers, slides, cover slips, glass rods, measuring cylinders (10 ml, 100 ml, 250 ml and 1000 ml) were used.

3.1.2 Equipments

During present investigations, following laboratory equipments viz., Autoclave, Hot Air Oven, Laminar Air Flow, BOD incubator (Remi), Research microscope, stereoscopic microscope, digital weighing balance (Wensar HBT 516), Centrifuge machine (Eppendorf 5810R), PCR machine (Eppendorf GR), Gel electrophoresis (Genexy, Scie-Plas), Gel Documentation Unit, vortex (Spinix), Deep freezer -20°C (Blue Star), freezer, digital camera.

3.1.3 Seeds

The chickpea genotype used in the study was JG-62 obtained from Pulse Research Unit, Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

3.1.4 Miscellaneous material

Inoculation needle, single and double distilled water, spirit lamp, spirit, scalpel, cotton, plastic pots, dissection needle, tray, rubber bands, mortar pestles, scissors etc. were used in present study.

3.1.5 Culture Media

Potato Dextrose Agar (PDA) and Potato Dextrose Broth (PDB) media were used during laboratory studies of fungus.

3.1.6 Chemicals

Different materials were used for study of molecular variability of *Rhizoctonia bataticola* isolates are given in Table 1.

Table 1. Material used for detection of molecular variability of *Rhizoctonia bataticola*

Sr. No.	Name of chemicals
1. DNA Extraction	
1	B-mercaptoethanol
2	Boric acid
3	Cetyl Trimethyl Ammonium Bromide (CTAB)
4	Chloroform
5	Ethyline Diamine Tetra Acetic Acid Disodium Salt (EDTA Na ₂)
6	Ethanol
7	Isoamyl alcohol
8	Isopropanol
9	Phenol
10	Sodium Chloride
11	Tris-HCl
2. PCR Amplification	
12	10X <i>Taq</i> buffer
13	MgCl ₂ 25 Mm
14	<i>Taq</i> DNA polymerase (5U/μl)
15	Template DNA

16	dNTP mix (10 mM)
17	Oligonucleotide primers
18	Sterile double distilled water
3. Poly acrylamide gel electrophoresis	
19	Acrylamide
20	NN-Bis-acrylamide
21	EDTA Na ₂
22	Methanol
23	Sodium hydroxide
24	APS
25	Gel loading dye 6X
26	Tris – HCl
27	Boric acid
28	Acetic acid glacial
29	Formaldehyde
30	Silicon grease
31	TEMED
32	DNA ladder 100 bp
33	Silver nitrate
4. Agarose gel electrophoresis	
34	Agarose
35	Boric acid
36	EDTA Na ₂
37	Tris base HCl
38	Methanol
39	Ethidium bromide
40	Tris buffer
41	DNA ladder 1 kb
5. Software's used	
42	AlphaEaseFC
43	NTSYS 2.02

3.1.7 Different solutions, buffers and reagents used for study

Different solutions, buffers and reagents used to perform present study are listed below along with their composition.

3.1.7.1 Extraction of DNA

- Extraction buffer [2 % w/v CTAB, 1.4 M NaCl, 20 mM EDTA, 100 mM Tris-HCl, 0.4 % β -mercaptoethanol (added at the time of use)]. (The extraction buffer was autoclaved before addition of β -mercaptoethanol)
- Chloroform : Isoamyl alcohol (24:1)
- 100 % Isopropanol (Ice-cold)
- Wash buffer : 70 % ethanol
- TE buffer (10mM Tris and 1mM EDTA, p^H 8.0) Autoclaved before use.
- 5 M NaCl (solution was sterilized by autoclaving).
- 1 M Tris HCl (p^H 8.0)
- 0.5 M EDTA Na₂ (p^H 8.0)

3.1.7.2 PCR Amplification

Various reagents used for the RAPD, SSR and ITS amplification are listed below:

- 10 picomole/ μ l Oligonucleotide RAPD, SSR and ITS primers
- 5U/ μ l *Taq* DNA polymerase
- 10 X *Taq* buffer
- 25 mM MgCl₂
- 10mM dNTP

3.1.7.3 Agarose gel electrophoresis

- 10 X TBE (0.9 M Tris base, 0.9 M Boric acid, 0.10 M EDTA)
- 6X Gel loading dye (10 mM Tris-Cl p^H 7.6, 0.03 % bromophenol blue, 0.03 % xylene cyanol FF, 60 % glycerol in water).
- Ethidium bromide (0.5 mg/ml)

3.1.7.4 Polyacrylamide gel electrophoresis

- 10 x TBE (0.9 M Tris base, 0.9 M Boric acid, 0.10 M EDTA)
- 10% Polyacrylamide Gel (PAGE) (Urea 100 g, Acrylamide 95 g, Bis acrylamide 5 g, 10x TBE 100 ml for total volume of 1 lit)
- 10% APS (0.1 g Ammonium per sulphate dissolved in 1 ml distilled water)

3.2 Methods

3.2.1 Preparation of media

The cultures of the isolates of *Rhizoctonia bataticola* thus obtained from the Akola, Nagpur, Amravati, Satara, Sangli and Kolhapur were maintained on Potato Dextrose Agar (PDA) media was used.

3.2.1.1 Potato Dextrose Agar media

The Potato Dextrose Agar (PDA) media was used for maintenance of cultures of *Rhizoctonia bataticola* and further morphological studies.

3.2.1.1.1 Procedure for preparation of PDA media

The composition of PDA media is given in Table 2. Potatoes were peeled and cut into small slices and gently washed. 200 gm of sliced potato added into 500 ml of water and allowed to boil till it gets soft. These boiled potatoes slices passed through muslin cloth, filtrate was collected into 2000 ml capacity beaker and final volume was made to 1000 ml by adding distilled water. The filtrate was allowed to boil, gently added 20 gram of dextrose followed by 20 gram of agar-agar with intermediate shaking for 3-5 minutes. The medium thus prepared was poured in 500 ml flask and plugged by non absorbent cotton and wrapped with paper over it. The PDA media was autoclaved at for 15 lbs/in² (121.6⁰C) for 15 minutes.

Table 2. The composition of Potato Dextrose Agar (PDA) media

Sr. No.	Ingredient	Quantity
1	Peeled and sliced potato	200 gm
2	Dextrose (C ₆ H ₁₂ O ₆)	20 gm
3	Agar-agar	20 gm
4	Distilled Water	1000 ml

3.2.2.1.2 Preparation of PDA plates and slants

The autoclaved petri plates and slants were transferred to laminar air flow chamber and kept under UV light for 20 minutes for maintaining aseptic condition. The PDA medium prepared in the flask was liquefied by keeping it into microwave oven for three minutes. Streptomycin @ 2.5 mg/l was added in lukewarm media and gently shaken for uniform mixing. 10 ml of lukewarm PDA was poured immediately into each petri plate approximately. These plates were kept for 24 hour in incubator to confirm free from contamination. Similarly warm liquefied PDA containing streptomycin was poured into slants. These slants were kept tilted for solidification and used to preserve purified culture of *Rhizoctonia bataticola* isolates.

3.2.3 Preparation of Potato Dextrose Broth (PDB)

It was prepared for mass multiplication of *Rhizoctonia bataticola* for extraction of DNA.

The peeled potatoes slices were boiled in 800 ml of distilled water and the extract was collected by filtering through a muslin cloth. The filtrated potato extract was transferred into another jar and 20 gram of dextrose was added to the mixture. The volume was made up to 1000 ml with distilled water. 100 ml broth was added into 250 ml flasks and autoclaved at 121.6⁰C for 15 minutes at 15 lbs/in². The detail composition of PDB is given in Table 3.

Table 3. The composition of Potato Dextrose Broth (PDB)

Sr. No.	Ingredient	Quantity
1	Peeled and sliced potato	200 gm
2	Dextrose (C ₆ H ₁₂ O ₆)	20 gm
3	Distilled Water	1000 ml

3.2.4 Isolates**Table 4. List of isolates**

Sr. No.	District	Location	Isolates No.
1	Akola	Akola	Rb-1
2	Nagpur	Nagpur	Rb-2
3	Amravati	Amravati	Rb-3
4	Satara	Umbraj	Rb-4
5	Satara	Karanje	Rb-5
6	Sangli	Sangli	Rb-6
7	Sangli	Kurundwad	Rb-7
8	Kolhapur	Kini	Rb-8

3.2.5 Identification of pathogen

The identification of *Rhizoctonia bataticola* isolates was done on the basis of morphological characters described by Aghakhani and Dubey, 2009 as well as on molecular basis by using ITS-1 and ITS-4 molecular markers (Aghakhani and Dubey, 2009^b).

Table 5. List of ITS primers with their sequence used for identification of *Rhizoctonia bataticola*

Sr. No.	Oligo Name	Primer Sequence (5'-3')
1	ITS1	TCCGTAGGTGAACCTGCGG
	ITS4	TCCTCCGCTTATTGATATGC

3.2.6 Maintenance of the culture

The culture of different isolates of *Rhizoctonia bataticola* were maintained on PDA at room temperature by adopting subsequent sub culturing at regular intervals. The seven days old culture was used for further studies.

3.2.7 Pathogenicity test

The pathogenicity test of eight isolates of *Rhizoctonia bataticola* isolated from the different parts of the Maharashtra region was tested by using cultivar JG-62. The inoculum of *Rhizoctonia bataticola* was prepared in sand maize meal media (1 part partially broken maize grain + 3 part sand + distilled water to moisten the media). The media was prepared by mixing broken maize grains with clean sand in plastic tub followed by moistening with distilled water. About 100 g mixture was transferred in 250 ml Erlenmayer flask, plugged using non absorbent cotton and sterilized at 15 p.s.i. for 30 minutes. The flasks containing the sterilized media were inoculated with mycelial disc of *Rhizoctonia bataticola* and incubated at $27 \pm 2^{\circ}\text{C}$ for 15 days. The inoculum was used for soil inoculation at 25 g kg^{-1} soil in pot experiments. 10 seeds of cultivar JG-62 were sown in each pot and with 3 replications. The per cent emergence and post-emergence mortality at 30 DAS was recorded and total mortality was calculated. In case of control, chickpea seeds were sown in pots containing uninoculated soil. Plants were observed periodically upto initiation of root rot symptoms and disease incidence (%).

The pathogenicity test was confirmed by proving Koch's Postulate and pathogens were confirmed as *Rhizoctonia bataticola*.

3.2.8 Grouping of isolates

The isolates of *Rhizoctonia bataticola* were tentatively divided in five groups on the basis of virulence as follows.

Category		Per cent root rot
I)	Non pathogenic	0%
II)	Weakly pathogenic	1-20%
III)	Moderately pathogenic	21-50%
IV)	Strongly pathogenic	51-70%
V)	Highly pathogenic	> 70%

(Pawar, 2010)

3.2.9 Molecular variability analysis

In present investigations, 8 isolates of *Rhizoctonia bataticola* were used to detect variations using Random Amplified Polymorphic DNA (RAPD), Simple Sequence Repeat (SSR) and Internal Transcript Spacer (ITS) markers.

3.2.10 Preparation of mycelium mat

The pure culture of eight *Rhizoctonia bataticola* isolates were mass multiplied by transferring mycelium tissue into 100 ml of PDB and kept for incubation inside BOD incubator at $27 \pm 2^{\circ}\text{C}$ for next seven days without disturbing the flasks. The mycelium mat from broth was transferred to blotting paper and allowed to air dry for 20 min at room temperature. This mycelial mat was used for isolation of genomic DNA.

3.2.11 Extraction of DNA

Genomic DNA was isolated from the eight selected isolates by the cetyl trimethyl ammonium bromide (CTAB) method (Murray and Thompson, 1980) with some modifications.

3.2.11.1 Reagents required

1.1M Tris HCl (p^H 8.0)

30.285 gram tris HCl dissolved in 150 ml of distilled water and volume was made up to 250 ml with distilled water. The buffer was autoclaved and stored at room temperature.

2. 0.5M Na₂ EDTA

37.224 gm dissolved Na₂ EDTA in 150 ml of distilled water by adding NaOH pellets till the p^H of the buffer came to 8.0 and the total volume of buffer was made up to 200 ml with distilled water. The buffer was autoclaved and stored at room temperature.

3. Extraction buffer

Tris HCl 50 Mm (p^H 8.0), NaCl 500 mM, Na₂EDTA 50 mM, p^H 8.0 and 1 per cent (v/v) β-Mercaptoethanol (Added immediately before use).

4. Chloroform : Isoamyl alcohol (24:1)

480 ml chloroform and 20 ml of isoamyl alcohol was mixed together.

5. Isopropanol

6. 70 per cent Ethanol

7. TE 10:1, p^H 8.0 (10 mM Tris; 1mM Na₂ EDTA p^H 8.0)

2 ml 1M Tris, p^H 8.0 and 0.4 ml of 0.5 M Na₂ EDTA, p^H 8.0 was added to 197.6 ml of sterile distilled water.

3.2.11.2 Procedure for isolation of fungal genomic DNA

The seven days old mycelium mat was transferred on sterilized blotter paper, air dried to remove moisture and media adhering to the mat. The dried mycelium mat was used for DNA isolation. Approximately, one gram of air dried fungal mat was quickly frozen in liquid nitrogen (-196⁰C) and crushed into powder form with the help of

sterilized mortar and pestle. The powder was immediately homogenized by adding pre-warmed (65⁰C) extraction buffer and transferred to two ml eppendorf tubes. The content of tube was shaken vigorously for one minute for uniform mixture of suspension. These tubes were incubated in a water bath at 65⁰C for one hour with gentle shaking at every 10 minutes. Tubes containing homogenate were centrifuged at 12000 rpm for 15 min at room temperature. Then supernatant was transferred into new 2 ml eppendorf tubes without disturbing the pellet of cell debris. Then 1 ml of chloroform: Isoamyl alcohol (24:1) was added and mixed gently but thoroughly to emulsify both the components for 5 min. Centrifugation was carried out at 12000 rpm for 15 min at room temperature.

The upper aqueous phase was transferred into new 2 ml eppendorf tubes with a wide bore pipette. Equal volume of ice-cold isopropanol was added and mixed by inversions. CTAB-DNA complexes formed a fibrous network. After mixing with isopropanol, the samples were kept at 4⁰C for 10 minutes and then centrifuged at 12000 rpm at room temperature for 10 min. After centrifugation a pellet was formed at the bottom of the eppendorf tubes. The supernatant was removed and the pellet was washed with 70% ethanol twice. The pellet was air-dried for 30-60 minutes and then dissolved in 100 µl of TE buffer. The pellets were allowed to dissolve completely overnight at 4⁰C without agitation. Then stored at -20⁰C until needed.

3.2.11.3DNA quantification

The DNA obtained after extraction was confirmed by running it on 0.8% agarose gel containing ethidium bromide at 0.5 mg/ml in a horizontal gel electrophoresis system. 6 µl of genomic DNA of each isolate + 3µl loading dye + 5µl sterile water loaded in each well. After completion of 5 cm run, the gel was observed under UV light and the DNA yield and quality was confirmed.

3.2.11.4 Genetic diversity analysis by Random Amplified Polymorphic DNA (RAPD)

The 15 RAPD primers were selected to study genetic diversity among *Rhizoctonia bataticola* isolates. The detailed of RAPD primers given in Table 6.

Table 6. List of RAPD primers used with their sequences

Sr. No.	Primer Screened	Sequence	Annealing Temp. (°C)
1	OPA-2	TGCCGAGCTG	34
2	OPA-5	AGGGGTCTTG	32
3	OPA-6	GGTCCCTGAC	34
4	OPA-7	GAAACGGGTG	32
5	OPA-9	GGGTAACGCC	34
6	OPA-10	GTGATCGCAG	32
7	OPA-13	CAGCACCCAC	34
8	OPA-17	GACCGCTTGT	32
9	OPB-11	GTAGACCCGT	32
10	OPB-14	TCCGCTCTGG	34
11	OPF-1	ACGGATCCTG	32
12	OPF-2	GAGGATCCCT	32
13	OPF-3	CCTGATCACC	32
14	OPF-4	GGTGATCAGG	32
15	OPF-5	CCGAATTCCC	32

The PCR was carried out in small reaction tubes 200 µl containing a reaction volume typically of 12.5 µl (Table 7 and 8) that was inserted into a thermal cycler (Eppendorf) that heated and cooled the reaction tubes within it to the precise temperature required for each step of the reaction. PCRs were run on the programmable thermal cycler given in Table 9.

Table 7. PCR reaction mix for 1x of 12.5 µl reaction

Sr. No.	Master Mix	1x
1	10x <i>Taq</i> buffer	1.25 µl
2	MgCl ₂ (25mM)	1.25 µl
3	dNTPs (10mM)	0.3 µl
4	<i>Taq</i> polymerase (5U/µl)	0.3 µl
5	Sterile distilled water	5.40 µl
	Total Volume	8.5 µl

Table 8. Constituents of PCR reaction

Sr. No.	PCR Reaction	Quantity
1	Master Mix vol.	8.5 µl
2	Primer	2.0 µl
3	Template DNA (37.5 ng)	2.0 µl
	Total Reaction Volume	12.5 µl

Table 9. Steps used for PCR- RAPD reaction

Name of step	Temperature (°C)	Time
Initial Denaturation	94 ⁰ C	1 min
- Denaturation 35 cycles - Annealing - Extension	94 ⁰ C	1 min
	37 ⁰ C	1 min
	72 ⁰ C	2 min
Final Extension	72 ⁰ C	7 min

After completion of the cycles the samples were kept at 4⁰C till electrophoresis.

3.2.11.5 Separation of RAPD amplified products by agarose gel electrophoresis

Electrophoresis of RAPD-PCR amplified analysis on agarose gel was carried out in 1x TBE buffer in horizontal gel electrophoresis. Cleaned and dried electrophoresis assembly was used for gel

electrophoresis. The gel tray was wiped and cleaned with methanol. It was set to prepare the gel 1 to 1.5% agarose gel solution was prepared in 1x TBE buffer. Ethidium bromide was added to the lukewarm liquefied gel solution as staining agent. PCR products were loaded in the well on the gel along with the 1 µl ladder as 10000bp marker in the first well. 3 µl 6x loading dye was loaded along with each PCR product in the wells. Then the gel was run for two hours at 80 volts. After completion of 10 cm run, the gel was observed and captured under UV light with the help of Bio-Rad Gel documentation system.

3.2.11.6 Genetic diversity analysis by Simple Sequence Repeat (SSR)

Six SSR primers were selected to study genetic diversity among *Rhizoctonia bataticola* isolates. The detailed of SSR primers given in Table 10.

Table 10. List of SSR primers used with their sequences

Sr. No.	Primer Name	Primer sequence (5'-3')	SSR Motif	Annealing Temperature (°C)
1	MB-2	F: TGCTGTGTATGGATGGATGG R: CATGGTCGATAGCTTGTCTCAG	(GT) ₁₁ (GA) ₆	57.0
2	MB-5	F: ACTTGGAGGAAATGGGCTTC R: GGATGGCGTTTAATAAATCTGG	(TG) ₉	54.0
3	MB-9	F: TGGCTGGGATACTGTGTAATTG R: TTAGCTTCAGAGCCCTTTGG	(CA) ₉	51.0
4	MB-14	F: CGTCTCTGAACCACCTTCATC R: TTCCTCCGTCCATCCTGAC	(CCA) ₅	57.0
5	MB-17	F: ACTGATTCACCGATCCTTGG R: GCTGGCCTGACTTGTTATCG	(CA) ₂₁	57.0
6	MB-18	F: GGTAGGAAATGACGAAGCTGC R: TGAGCACTCTAGCACTCCAAAC	(CAACA) ₆	57.0

The PCR was carried out in small reaction tubes, containing a reaction volume typically of 12.5 μl (Table 11 and 12) that was inserted into a thermal cycler (Eppendorf tube 200 μl) that heated and cooled the reaction tubes within it to the precise temperature required for each step of the reaction. PCRs were run on the programmable thermal cycler given in Table 13.

Table 11. PCR reaction mix for 1x of 12.5 μl reaction

Sr. No.	Master Mix	1x
1	10 x <i>Taq</i> buffer	1.25 μl
2	MgCl ₂ (25 mM)	1.25 μl
3	dNTPs (10mM)	0.3 μl
4	<i>Taq</i> polymerase (5U/ μl)	0.3 μl
5	Sterile distilled water	5.4 μl
	Total Volume	8.5 μl

Table 12. Constituents of PCR reaction

Sr. No.	PCR Reaction	Quantity
1	Master Mix vol.	8.5 μl
2	Primer (Forward)	1.0 μl
3	Primer (Reverse)	1.0 μl
4	Template DNA (37.5 ng)	2.0 μl
	Total Reaction Volume	12.5 μl

Table 13. Steps used for PCR-SSR

Name of step	Temperature ($^{\circ}\text{C}$)	Time (Min.)
Initial Denaturation	94	5
40 cycles { - Denaturation - Annealing - Extension	94	45 Sec.
	57 *	45 Sec.
	72	2
Final Extension	72	10

* Annealing temperature varied from primer to primer given in the Table 13.

After completion of the cycles keep the samples at 4⁰C till electrophoresis.

3.2.11.7 Internal Transcribed Spacer (ITS) amplification

Genetic variability of *Rhizoctonia bataticola* was evaluated by using ITS primers (Aghakhani and Dubey, 2009^b) (Table 14).

Table 14. List of ITS primers used with their sequences

Sr. No.	Oligo Name	Primer Sequence (5'-3')
1	ITS 1	TCCGTAGGTGAACCTGCGG
	ITS 4	TCCTCCGCTTATTGATATGC

3.2.11.8 Procedure for PCR reaction

Sterile PCR tubes were numbered and placed on PCR tube stand. At first 2 µl of DNA was added to each PCR tube followed by master mix given in table 11 and 12. The samples were mixed by brief centrifugation to bring down the content of tube. PCRs were run on the programmable thermal cycler given in table 15. PCR products were separated by electrophoresis in 1 to 1.5 per cent agarose gels run in 1x TBE, stained with ethidium bromide and visualized with a UV transilluminator.

Table 15. Steps used for PCR-ITS reaction

Name of step	Temperature (°C)	Time
Initial Denaturation	94 ⁰ C	5 min.
35 cycles { - Denaturation - Annealing - Extension	94 ⁰ C	1 min.
	55 ⁰ C	1 min.
	72 ⁰ C	30 sec.
Final Extension	72 ⁰ C	10 min.

3.2.12 Separation of SSR - PCR amplified products by Polyacrylamide Gel Electrophoresis (PAGE)

3.2.13 Composition of Solution, Chemicals and Reagents

1.10% Polyacrylamide Gel (for total volume 1 lit)

Urea	100 g
Acrylamide	95 g
Bis acrylamide	5 g
10X TBE	100 ml

Total volume was made upto 1 lit with distilled water and 0.085% of 10% Ammonium Per Sulphate solution and 0.075% of TEMED was added before use.

Solutions used for staining of gel after electrophoresis of SSR-PCR product were as follows,

1. Staining solution (1 lit)

Methanol	100 ml
Glacial Acetic Acid	5000 μ l
Silver nitrate	6 g

Total volume was made upto 1 lit with double distilled water.

2. Fixing solution (1 lit)

Methanol	100 ml
Glacial Acetic Acid	5000 μ l

Total volume was made upto 1 lit with double distilled water.

3. Developing solution (1 lit)

Sodium hydroxide	30 g
Formaldehyde	3000 μ l

Total volume was made upto 1 lit with double distilled water.

3.2.14 Electrophoresis of SSR-PCR product

The SSR analysis was carried out in vertical gel electrophoresis. The PCR products were separated electrophoretically in 10% polyacrylamide gel (PAGE). The cleaned and dried vertical gel electrophoresis assembly was used. The glass plates, spacer, combs etc. were wiped and cleaned with ethanol. The polyacrylamide gel solution was poured between glass plates and comb was placed onto the gel and allowed to solidify for one to one and half hours. 1X TBE buffer was used as tank buffer. After solidification of the gel, comb was removed and pre-running of the gel for 30-60 min was followed. After pre-running of gel, the SSR-PCR product (7-10 μ l) with 3 μ l loading dye each in a well was added along with 100bp DNA ladder (1 μ l) with 3 μ l loading dye in the first well. The gel was run for 3 hrs. at 70-80V and 250-300mA. After running of the DNA sample, gel was removed carefully from the unit and gel staining procedure was followed.

3.2.14.1 Staining of the gel

After electrophoresis the gels were carefully removed from the glass plates and transferred to a tray containing double distilled water and kept for 5 min for gentle shaking. The distilled water in the above tray was replaced with fixing solution and kept for another 5 min with gentle shaking. The fixing solution was removed from the tray and retained for further use. Staining solution was now poured in the tray. The staining was done for 2.5 min with gentle shaking. The staining solution was removed from the tray and gel was rinsed for a while in distilled water. The gel was transferred to a developing solution. The solution in the tray was shaking gently for 5 to 10 min to resolve the DNA bands. The staining was stopped by rinsing the gel for 5 min in the fixing solution. The gel was placed on the platform of the gel documentation system and photographed under UV converted white light.

3.2.15 Data analysis

The gel image of RAPD and SSR analysis were captured and visualized under light in gel documentation system (Biorad).

- i. Data was scored as the presence (1) or absence (0) of individual band for each isolate.
- ii. The similarity index was calculated as

$$\text{Similarity Index} = \frac{2 N_{AB}}{N_A + N_B}$$

Where,

N_{AB} : is the number of bands shared by individuals A and B and

N_A and N_B : Total number of bands in individual A and B, respectively.

The data was used to generate similarity coefficient using simple matching coefficient based on RAPD and SSR bands scoring. The similarity coefficient between each pair of accessions were then used to construct a dendrogram using the Unweighted Pair Group Method with Arithmetic Average (UPGMA).

CHAPTER IV

RESULTS AND DISCUSSION

The chickpea dry root rot caused by *Rhizoctonia bataticola* is an important fungal pathogen restricting chickpea production worldwide and causes the loss to the extent of 10-60 per cent. (Dhar and Chaudhary, 2001; Subramanian Sundravada *et al.*, 2012). The pathogen has high genetic variability therefore an investigation was undertaken to study genetic variability by using RAPD and SSR markers.

4.1 Pathogenicity

The pathogenicity test of 8 isolates of *Rhizoctonia bataticola* isolated from the different parts of the Maharashtra region was tested by using cultivar JG-62. The results are presented in Table 16.

The result indicates that amongst all the eight isolates of *Rhizoctonia bataticola* all proved to be pathogenic to cultivar JG-62. On the basis of per cent mortality these eight isolates were grouped as strongly pathogenic (2), moderately pathogenic (6).

Table16. Pathogenicity test of *Rhizoctonia bataticola* against variety JG-62

Sr. No.	Isolates	Total no. of seeds sown	Total no. of seeds germinated	No. of plants infected	Days to initiate symptoms	Per cent mortality due to pathogen (30 DAS)	Place
1.	Rb-1	30	25	11	24	44.00	Akola
2.	Rb-2	30	22	11	22	50.00	Nagpur
3.	Rb-3	30	26	15	25	57.69	Amravati
4.	Rb-4	30	27	13	26	48.14	Umbraj
5.	Rb-5	30	25	12	22	48.00	Karanje
6.	Rb-6	30	23	10	23	43.47	Sangli
7.	Rb-7	30	25	13	24	52.00	Kurundwad
8.	Rb-8	30	23	9	24	39.13	Kini
9.	Control	30	25	0	0	0	

Jayalakshmi *et al.* (2008) reported that pot culture experiment was conducted for screening 12 promising entries of chickpea (JG-11, GBS 964, GBS 963, GBC 2, GBM 6, GBM 2, GBM 10, BGD 103, GCP 107, GCP 101, A1, ICCV 10, L-550+1 susceptible check JG-62) against dry root rot during *rabi* season, 2004-05 and 2005-06. Out of 12 genotype tested against root rot disease, 4 resistant, 2 moderately susceptible, 3 susceptible and 3 highly susceptible to dry root rot disease were observed. The present study showed that all the 8 isolates of *Rhizoctonia bataticola* proved to be pathogenic to cultivar JG-62. Aghakhani and Dubey (2009^a) conducted pathogenicity test of twenty-three isolates of *Rhizoctonia bataticola* (*Macrophomina phaseolina*) causing dry root rot of chickpea collected from 10 different major chickpea growing states of India. Highly susceptible cultivar Pusa 212 was susceptible and all the isolates were pathogenic under artificially inoculated conditions.

4.2 Grouping of *Rhizoctonia bataticola* isolates based on their pathogenic ability

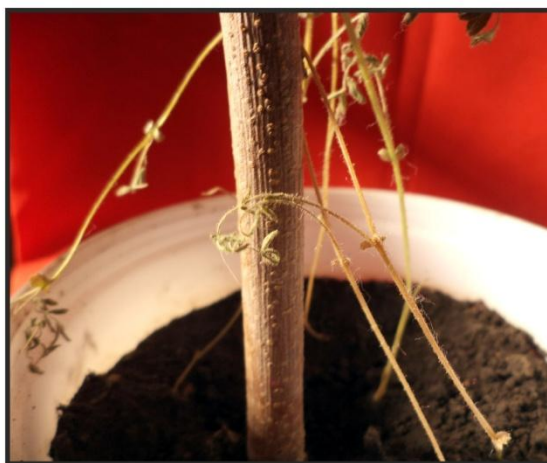
The data based on pathogenic variability in different isolates of *Rhizoctonia bataticola* are given in Table 16. The isolates of *Rhizoctonia bataticola* were tentatively divided in five groups based on their pathogenic ability on cultivar JG-62 (Table 17).

Category	Per cent mortality due to root rot
1. Non pathogenic	0 %
2. Weakly pathogenic	1-20 %
3. Moderately pathogenic	21-50 %
4. Strongly pathogenic	51-70 %
5. Highly pathogenic	>70

Grouping of *Rhizoctonia bataticola* isolates from different cultivated crops have been reported by Raut and Ingle (1989) and Aghakhani and Dubey (2009^a).



Pathogenicity experiment



Typical symptom of root rot of chickpea



**Strongly
pathogenic plant**



**Moderately
pathogenic plant**



Control

Plate 1. Grouping of *Rhizoctonia bataticola* isolates on the basis of pathogenicity

Table 17. Grouping of *Rhizoctonia bataticola* isolates on the basis of pathogenicity

Sr. No.	Category	Per cent mortality due to root rot	Isolates
1.	Non-pathogenic (NPI)	0	Nil
2.	Weakly pathogenic (WPI)	1-20	Nil
3.	Moderately pathogenic (MPI)	21-50	Rb-1, Rb-2, Rb-4, Rb-5, Rb-6, Rb-8
4.	Strongly pathogenic (SPI)	51-70	Rb-3, Rb-7
5.	Highly pathogenic (HPI)	>70	Nil

4.3 *Rhizoctonia bataticola* isolates selected for the molecular study

The isolates of *Rhizoctonia bataticola* were obtained from different districts of Maharashtra. The eight isolates viz., Akola, Nagpur, Amravati, Umbraj, Karanje, Sangli, Kurundwad and Kini were selected for the analysis. The *Rhizoctonia bataticola* specific ITS primer pair ITS-1 (TCCGTAGGTGAACCTGCGG) and ITS-4 (TCCTCCGCTTATTGATATGC) were used for molecular confirmation of the isolates. All the isolates of *Rhizoctonia bataticola* yielded the 500 bp band with the ITS marker, hence confirm the culture of *Rhizoctonia bataticola* were as per the result of Aghakhani and Dubey (2009^b).

4.4 RAPD primers selected for the study

The 15 RAPD primers supplied by “Genaxy” were used to evaluate the molecular variability in eight isolates of *Rhizoctonia bataticola*. The PCR (Polymerases Chain Reaction) amplified product of each primer were resolved on 1.2 % agarose gel electrophoresis and the size of the amplified product was compared with 1 kb DNA ladder.

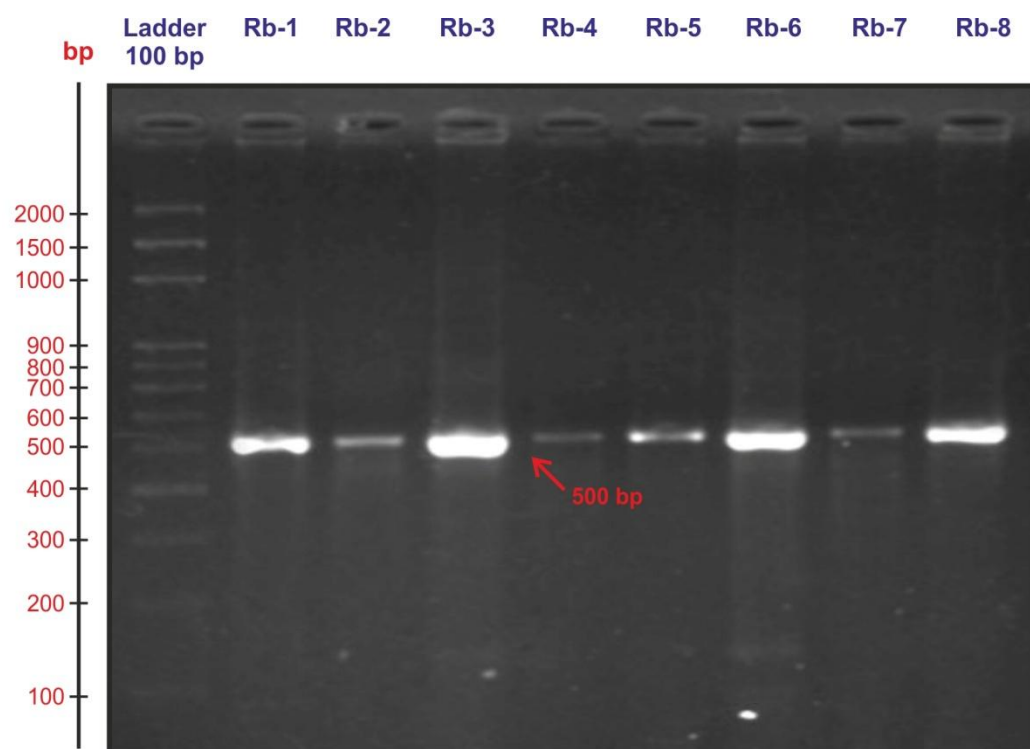


Plate 2. Molecular confirmation of *Rhizoctonia bataticola* showing 500 bp band size with ITS marker



Plate 3(a). Growth of *Rhizoctonia bataticola* isolates on PDA

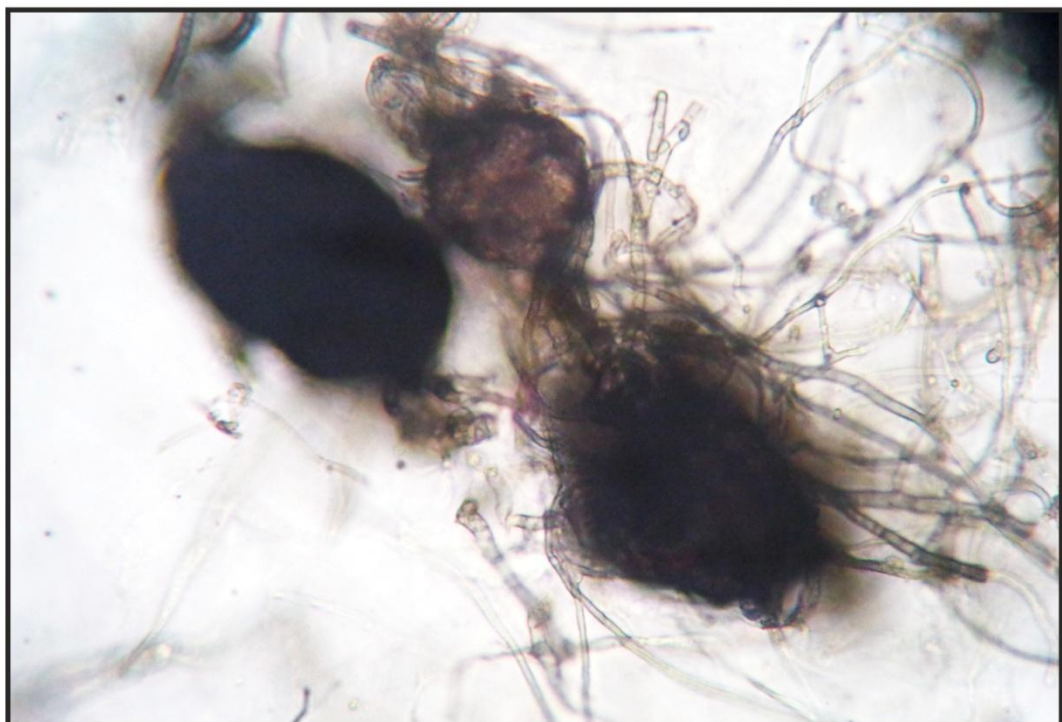


Plate 3(b). Sclerotia of *Rhizoctonia bataticola*

Out of 15 primers screened for RAPD during present study, 10 primers were polymorphic and primers OPA-6, OPF-1, OPF-2, OPF-3 and OPF-5 observed no banding pattern for the set of eight isolates of *Rhizoctonia bataticola* under study.

Table18. Per cent polymorphism observed in RAPD primers

Sr. No.	Primer	Bands	Polymorphic bands	% polymorphism
1	OPA-2	13	11	84.62
2	OPA-5	9	8	88.89
3	OPA-7	12	9	75
4	OPA-9	10	9	90
5	OPA-10	11	9	81.82
6	OPA-13	13	10	76.92
7	OPB-11	6	6	100
8	OPB-14	14	11	78.57
9	OPF-4	6	5	83.33
10	OPA-17	7	7	100
Total		101	85	84.16

4.5 RAPD banding pattern

The polymorphic banding pattern was observed in 10 primers. The genetic variability in *Rhizoctonia bataticola* was observed in the RAPD banding pattern and is given below.

The banding pattern observed in primer OPA-2 is presented in Plate 4. The primer amplified in 13 amplicons among eight isolates of *Rhizoctonia bataticola*. The size of the amplicons amplified ranges from 4860bp to 381.81bp. The polymorphism observed in this primer was 84.62 %. The details of the 13 RAPD bands types are as follows.

Band type 1 (4860bp)	: This type of band was observed in Nagpur and Kini isolates.
Band type 2 (2302.58bp)	: This type of band was observed in all isolates.
Band type 3 (2013.938bp)	: This type of band was observed in Akola, Nagpur, Karanje, Kurundwad and Kini isolates.
Band type 4 (1784.20bp)	: This type of band was observed in only kurundwad isolate.
Band type 5 (1442.88bp)	: This type of band was observed in isolates of Nagpur, Amravati, Umbraj, Karanje, Sangli and Kini.
Band type 6 (1202.93bp)	: This type of band was observed in the all isolates except Umbraj.
Band type 7 (954.11bp)	: This type of band was observed in the isolates of Akola, Nagpur, Sangli, Kurundwad and Kini.
Band type 8 (834.52bp)	: This type of band was observed in isolates of Nagpur and Kini.
Band type 9 (740.34bp)	: This type of band was observed in the isolates of Akola, Nagpur, Sangli, Kurundwad and Kini.
Band type 10 (628.44bp)	: This type of band was observed in all the isolates.
Band type 11 (569.88bp)	: This type of band was observed in the isolates of Akola and Nagpur.
Band type 12 (544.13bp)	: This type of band was observed in all isolates except Karanje.
Band type 13 (381.82bp)	: This type of band was observed in the isolates of Akola, Nagpur, Sangli, Kurundwad and Kini.

The banding pattern observed in primer OPA-5 is presented in Plate 5. The primer amplified 9 amplicons among eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 2714.82 bp to 691.59bp. The polymorphism observed in the primer was 88.89 %. The details of the 9 RAPD band types are as follows.

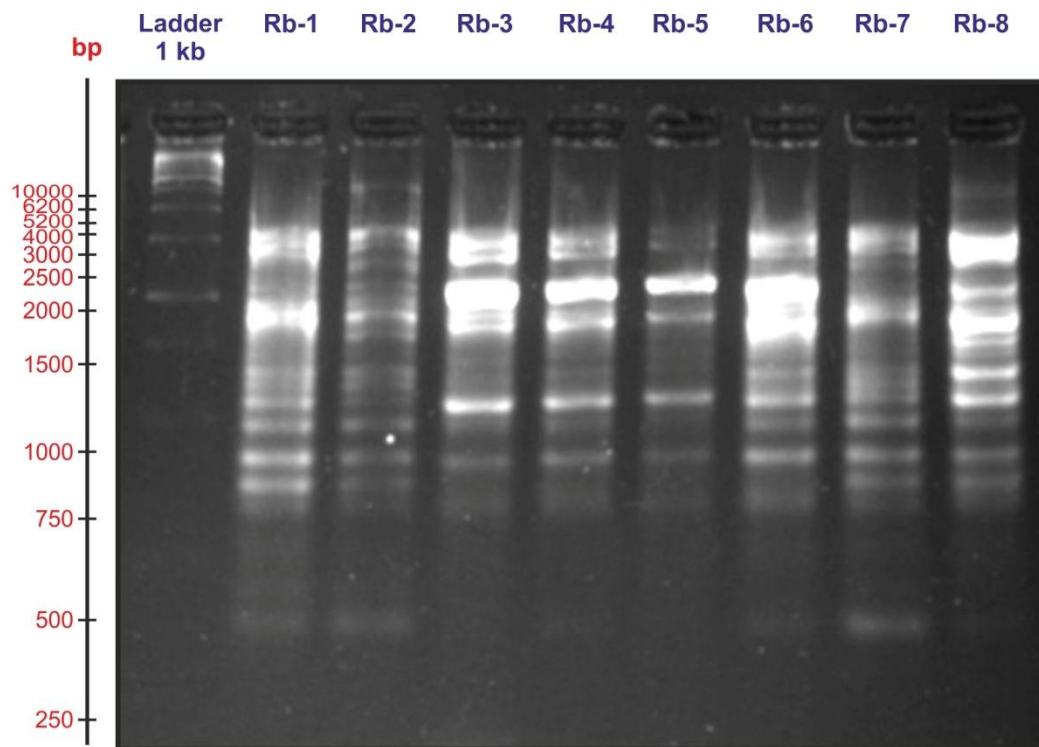


Plate 4. RAPD banding pattern of primer OPA-02

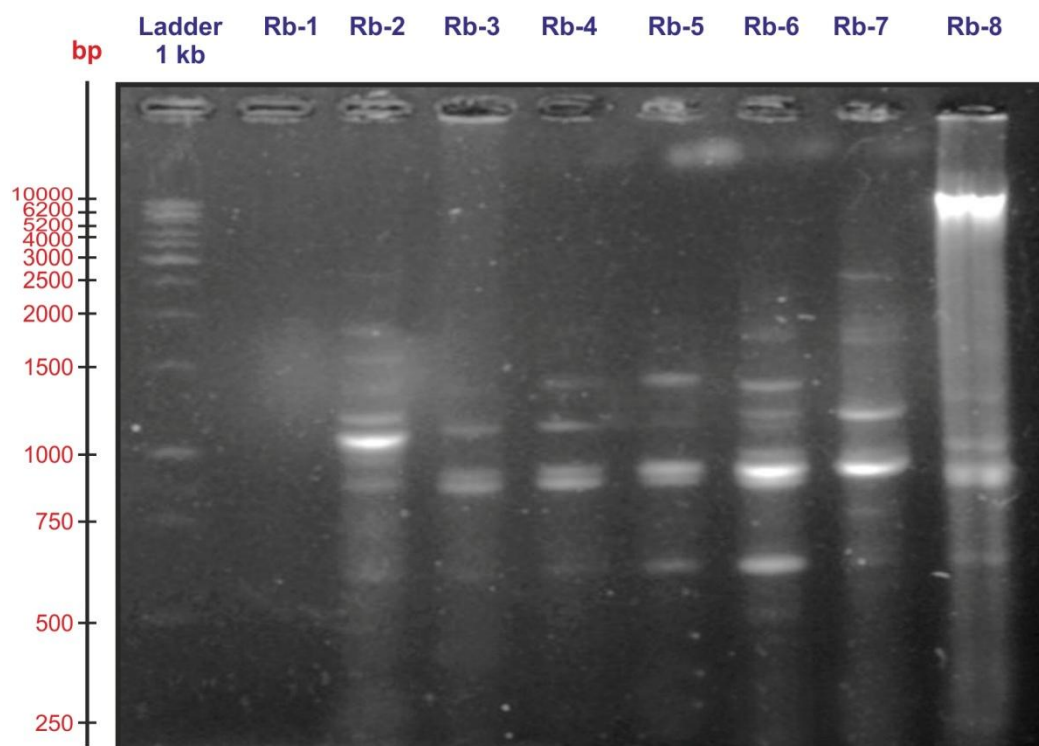


Plate 5. RAPD banding pattern of primer OPA-05

- Band type 1 (2714.82bp) : This type of band was observed only in the Kurundwad isolate.
- Band type 2 (1847.37bp) : This type of band was observed in isolates of Nagpur, Sangli and Kurundwad.
- Band type 3 (1624.46bp) : This type of band was observed only in isolate of Nagpur.
- Band type 4 (1409.13bp) : This type of band was observed in isolates of Umbraj, Karanje, Sangli.
- Band type 5 (1332.54bp) : This type of band was observed in the isolates of Nagpur, Sangli and Kurundwad.
- Band type 6 (1210.00bp) : This type of band was observed in the isolates of Nagpur, Amravati, Umbraj.
- Band type 7 (900bp) : This type of band was observed in the isolates of Nagpur, Sangli, Kurundwad, Kini.
- Band type 8 (860.07bp) : This type of band was observed in all isolates except Akola.
- Band type 9 (691.59bp) : This type of band was observed in the isolates of Umbraj, Karanje, Sangli and Kini.

The banding pattern observed in primer OPA-7 is presented in Plate 6. The primer amplified 12 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 4860.41bp to 500.88bp. The polymorphism observed in the primer was 75%. The details of the 12 RAPD band types are as follows.

- Band type 1 (4860.41bp) : This type of band was observed in isolates from Akola, Nagpur, Kurundwad and Kini.
- Band type 2 (3482.15bp) : This type of band was observed in isolates of Akola, Umbraj, Karanje, Kurundwad.
- Band type 3 (2176.09bp) : This type of monomorphic band was observed in all isolates.
- Band type 4 (1913.98bp) : This type of band was observed in isolates from Akola, Nagpur, Amravati and Kini.
- Band type 5 (1697.40bp) : This type of band was observed in isolates from Akola, Umbraj, Karanje, Sangli, Kurundwad.

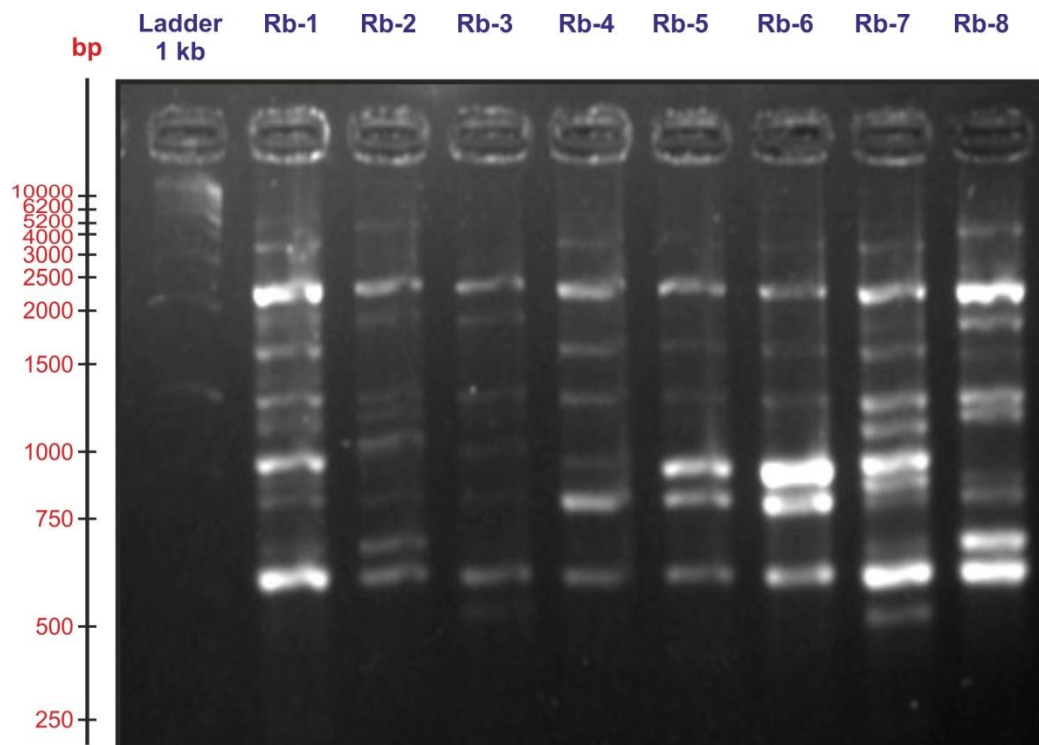


Plate 6. RAPD banding pattern of primer OPA-07

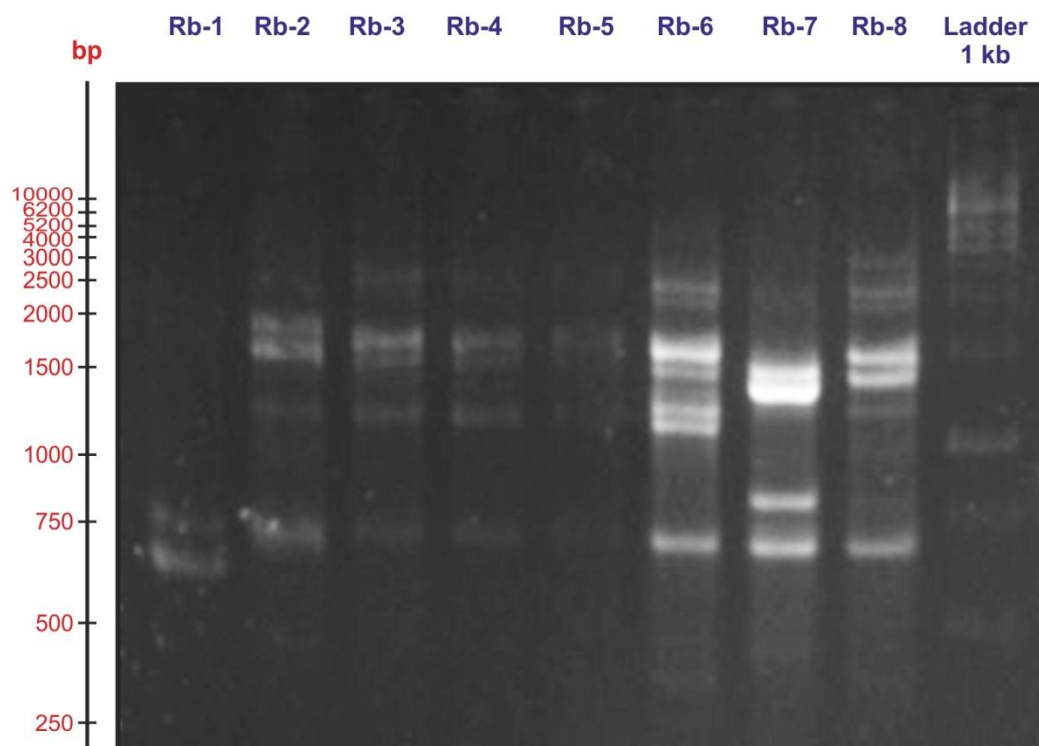


Plate 7. RAPD banding pattern of primer OPA-09

- Band type 6 (1402.93bp) : This type of band was observed in all isolates.
- Band type 7 (1108.19bp) : This type of band was observed in the isolates from Akola, Nagpur and Kini.
- Band type 8 (1026.00bp) : This type of band was observed in the isolates of Nagpur and Kurundwad.
- Band type 9 (890.74bp) : This type of band was observed in isolates of Umbraj, Karanje, Sangli, Kurundwad.
- Band type 10 (644.79bp) : This type of band was observed in isolates of Nagpur and Kini.
- Band type 11 (628.44bp) : This type of band was observed in all isolates.
- Band type 12 (500.88bp) : This type of band was observed in isolates from Amravati and Kurundwad.

The banding pattern observed in primer OPA-9 is presented in Plate 7. The primer amplified 10 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 2500bp to 644.93bp. The polymorphism observed in the primer was 90%. The details of the 10 RAPD band types are as follows.

- Band type 1 (2500bp) : This type of band as observed in isolates from Amravati, Umbraj, Karanje, Sangli and Kini.
- Band type 2 (2239bp) : This type of band was observed in isolates from Sangli and Kini.
- Band type 3 (1655bp) : This type of band was observed only in isolates from Nagpur, Amravati, Umbraj, Karanje, Sangli and Kurundwad.
- Band type 4 (1477.14bp) : This type of band was observed in isolates from Sangli, Kurundwad and Kini.
- Band type 5 (1279.93bp) : This type of band was observed only in the isolate from Kurundwad.
- Band type 6 (1171.30bp) : This type of band was observed in isolates from Nagpur, Amravati, Umbraj, Karanje, Sangli and Kini.
- Band type 7 (1034.75bp) : This type of band was observed only in isolate from Sangli.

- Band type 8 (775.84bp) : This type of band observed only in isolate of Kurundwad.
- Band type 9 (684.09bp) : This type of band was observed in all the isolates.
- Band type 10 (694.93bp) : This type of band was observed only in isolates from Akola.

The banding pattern observed in primer OPA-10 is presented in Plate 8. The primer amplified 11 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 2541.51bp to 412.04bp. The polymorphism observed in the primer was 81.82 %. The details of the 11 RAPD band types are as follows.

- Band type 1 (2541.5bp) : This type of band was observed in isolates from Nagpur, Amravati, Umbraj, Karanje and Sangli.
- Band type 2 (2329.62bp) : This type of band was observed in isolates of Nagpur, Umbraj and Karanje.
- Band type 3 (1577.66bp) : This type of band observed in all isolates.
- Band type 4 (1271.30bp) : This type of band was observed in isolates from Karanje, Sangli.
- Band type 5 (858.85bp) : This type of band was observed in isolate from Kini.
- Band type 6 (664.09bp) : This type of band was observed in all isolates.
- Band type 7 (626.68bp) : This type of band was observed only in the isolate of Akola.
- Band type 8 (519.81bp) : This type of band was observed in the isolates of Karanje, Sangli.
- Band type 9 (507.02bp) : This type of band was observed in isolates from Karanje, Sangli.
- Band type 10 (460.73bp) : This type of band was observed in isolates of Karanje, Sangli, Kurundwad.
- Band type 11 (412.04bp) : This type of band was observed in isolates from Umbraj, Karanje, Sangli.

The banding pattern observed in primer OPA-13 is presented in Plate 9. The primer amplified 13 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 2627.234

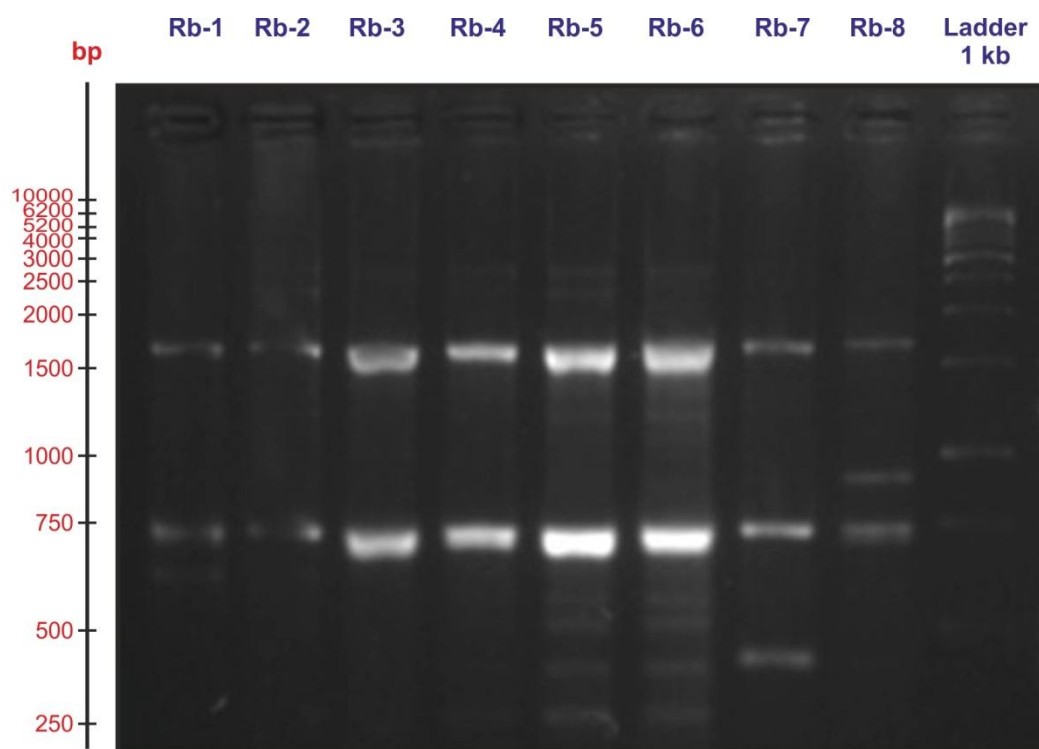


Plate 8. RAPD banding pattern of primer OPA-10

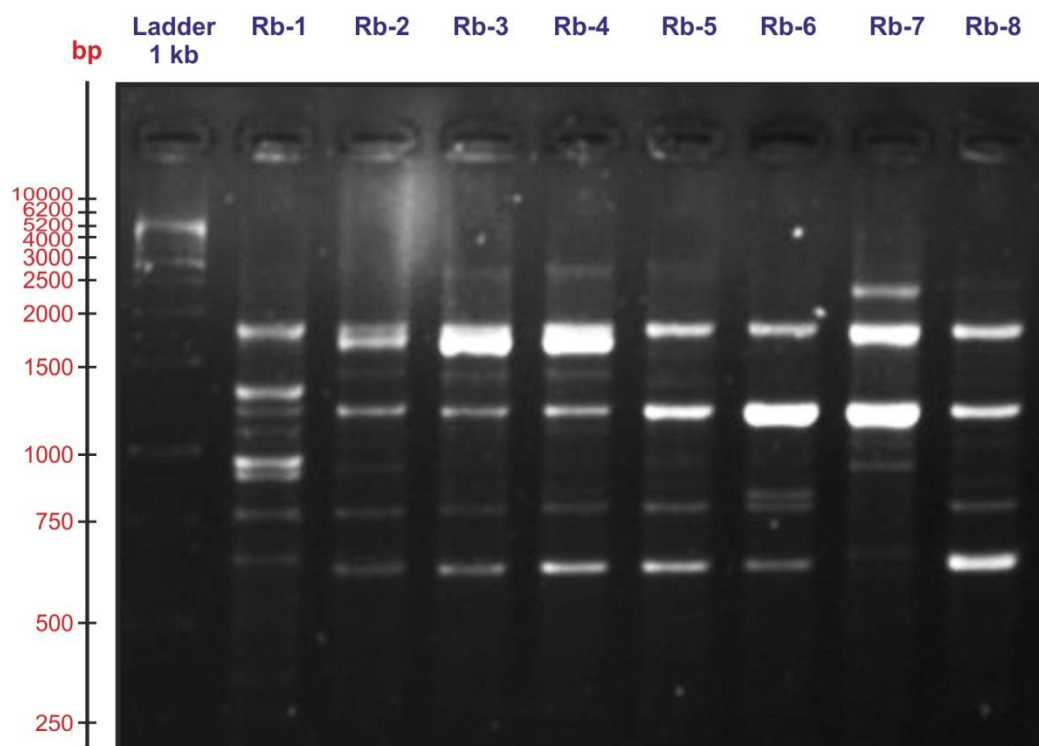


Plate 9. RAPD banding pattern of primer OPA-13

bp to 632bp. The polymorphism observed in the primer was 76.92 %. The details of the 13 RAPD band types are as follows.

- Band type 1 (2627.234bp) : This type of band was observed in isolates from Amravati, Umbraj, Karanje and Kurundwad.
- Band type 2 (2055.99bp) : This type of band was observed in the isolates from Kurundwad and Kini.
- Band type 3 (1784.40bp) : This type of band was observed in all isolates.
- Band type 4 (1678.67bp) : This type of band was observed in the isolates of Nagpur, Amravati, Umbraj.
- Band type 5 (1521.83bp) : This type of band was observed in the isolates of Nagpur, Amravati, Umbraj.
- Band type 6 (1391.87bp) : This type of band was observed only in isolate from Akola.
- Band type 7 (1334.43bp) : This type of band was observed in all isolates.
- Band type 8 (1148.44bp) : This type of band was observed in isolates from Akola and Kini.
- Band type 9 (909.51bp) : This type of band was observed in isolates from Akola, Kurundwad.
- Band type 10 (852bp) : This type of band was observed in isolates from Akola, Nagpur, Sangli.
- Band type 11 (810bp) : This type of band was observed only in isolate from Sangli.
- Band type 12 (770bp) : This type of band was observed in all isolates except Kurundwad.
- Band type 13 (632bp) : This type of band was observed in all isolates.

The banding pattern observed in primer OPB-11 is presented in Plate 10. The primer amplified 6 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 2159.78bp to 623bp. The polymorphism observed in the primer was 100 %. The details of the RAPD band types are as follows.

Band type 1 (2159.78bp)	: This type of band was observed in isolates of Amravati, Umbraj, Karanje, Sangli.
Band type 2 (1402.27bp)	: This type of band was observed in the isolates of Umbraj, Karanje, Kurundwad, Kini.
Band type 3 (1200bp)	: This type of band was observed in the isolate of Umbraj.
Band type 4 (1000bp)	: This type of band was observed in the isolate of Umbraj.
Band type 5 (782bp)	: This type of band was observed in isolate of Umbraj.
Band type 6 (623bp)	: This type of band was observed in isolate of Kurundwad.

The banding pattern observed in primer OPB-14 is presented in Plate 11. The primer amplified 14 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 1951.44bp to 336bp. The polymorphism observed in the primer was 78.75 %. The details of the 14 RAPD band types are as follows.

Band type 1 (1951.44bp)	: This type of band was observed only in isolate from Nagpur.
Band type 2 (1565.92bp)	: This type of band was observed only in the isolate of Sangli.
Band type 3 (1331.87bp)	: This type of band was observed in isolates from Nagpur, Sangli.
Band type 4 (1100bp)	: This type of band was observed in all the isolates.
Band type 5 (950bp)	: This type of band was observed in all the isolates.
Band type 6 (827.33bp)	: This type of band was observed in isolates from Amravati and Sangli.
Band type 7 (800bp)	: This type of band was observed in isolate from Sangli.
Band type 8 (790bp)	: This type of band was observed in isolate from Sangli.
Band type 9 (760bp)	: This type of band was observed only in isolate from Nagpur.

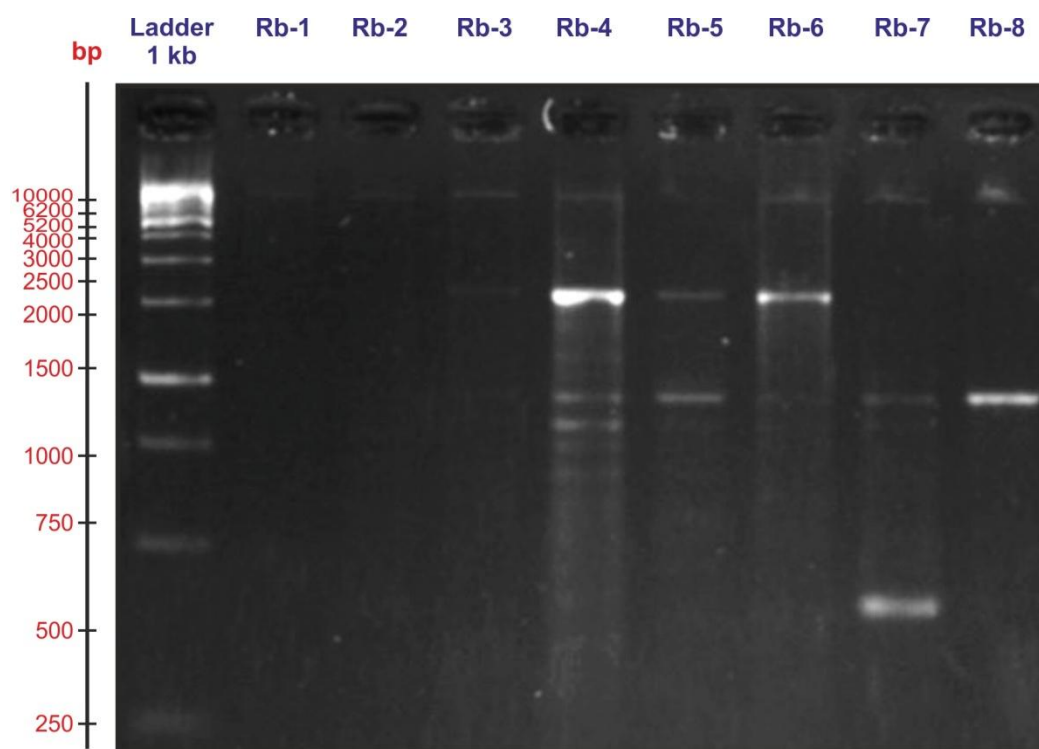


Plate 10. RAPD banding pattern of primer OPB-11

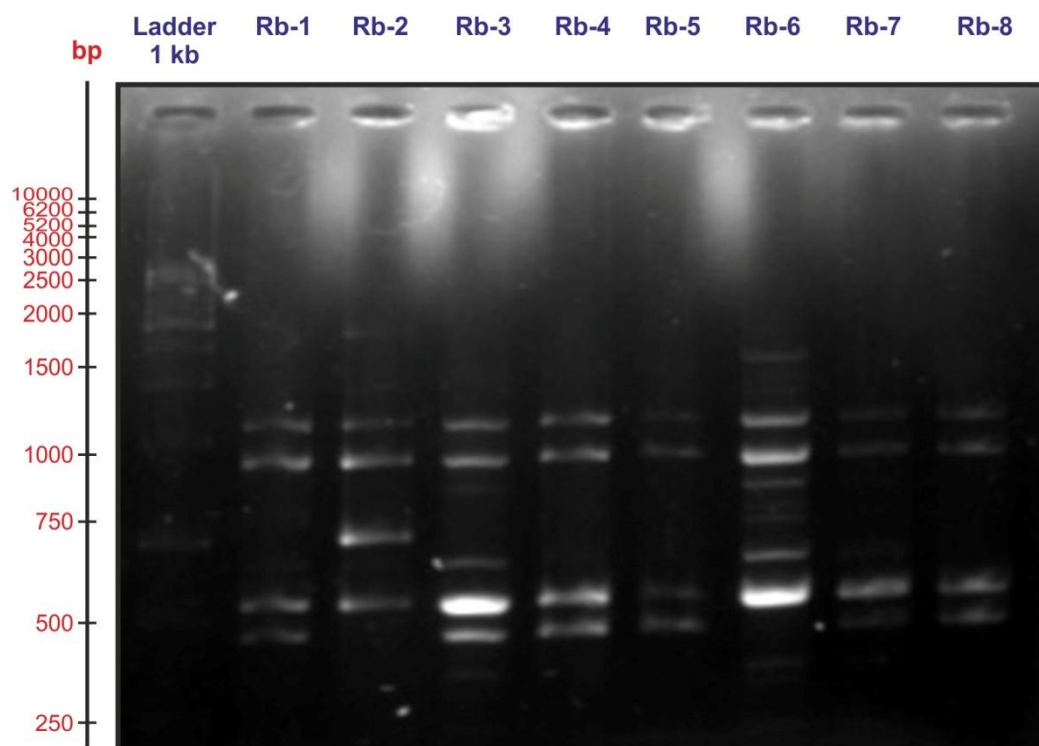


Plate 11. RAPD banding pattern of primer OPB-14

Band type 10 (657bp)	: This type of band was observed in isolates from Akola, Amravati, Sangli, Kurundwad.
Band type 11 (590bp)	: This type of band was observed in all isolates.
Band type 12 (487bp)	: This type of band was observed in isolates from Akola, Amravati, Umbraj, Karanje, Kurundwad, Kini.
Band type 13 (400bp)	: This type of band was observed in isolates from Amravati, Sangli.
Band type 14 (336bp)	: This type of band was observed only in isolate from Sangli.

The banding pattern observed in primer OPF- 4is presented in Plate 12. The primer amplified 6 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 950bp to 262.17bp. The polymorphism observed in the primer was 83.33 %. The details of the 6 RAPD band types are as follows.

Band type 1 (950bp)	: This type of band was observed in isolates of Umbraj, Karanje, Kurundwad, Kini.
Band type 2 (888.84bp)	: This type of band was observed in the isolates of Akola, Umbraj, Karanje, Kurundwad, Kini.
Band type 3 (589.41bp)	: This type of band was observed in the isolates of Amravati, Umbraj, Karanje, Kini.
Band type 4 (432.89bp)	: This type of band was observed only in isolates of Akola.
Band type 5 (349.17bp)	: This type of band was observed in all isolate except Amravati.
Band type 6 (262.17bp)	: This type of band was observed in all isolates.

The banding pattern observed in primer OPA-17is presented in Plate 13. The primer amplified 7 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 890bp to 550bp. The polymorphism observed in the primer was 100%. The details of the 11 RAPD band types are as follows.

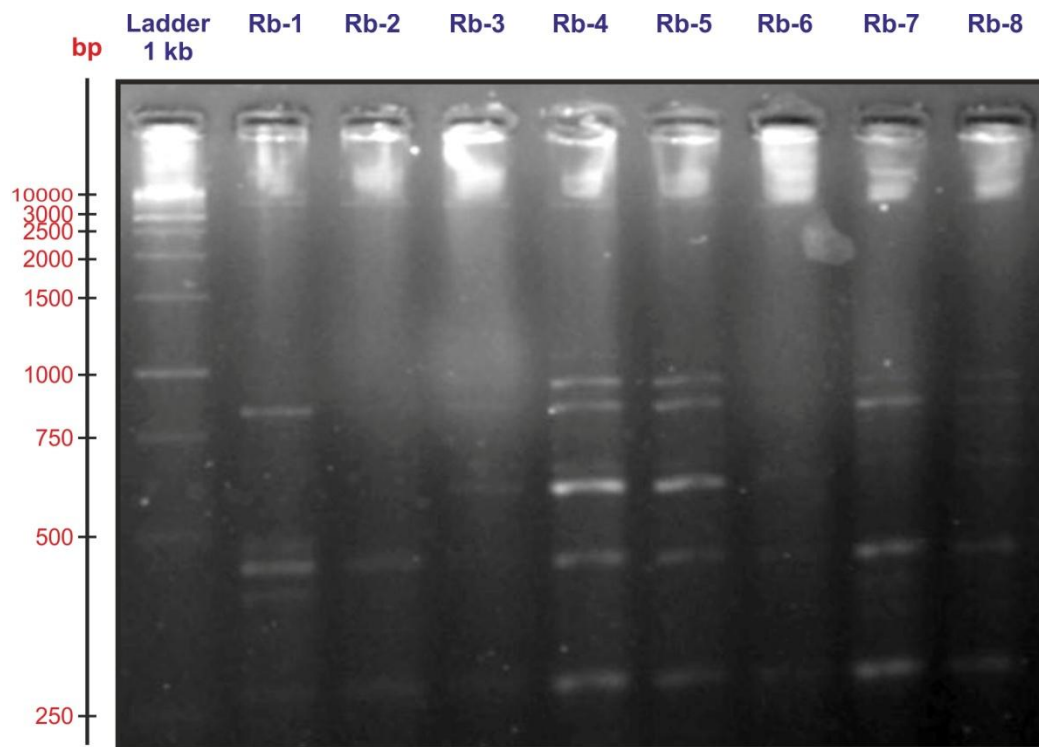


Plate 12. RAPD banding pattern of primer OPF-4

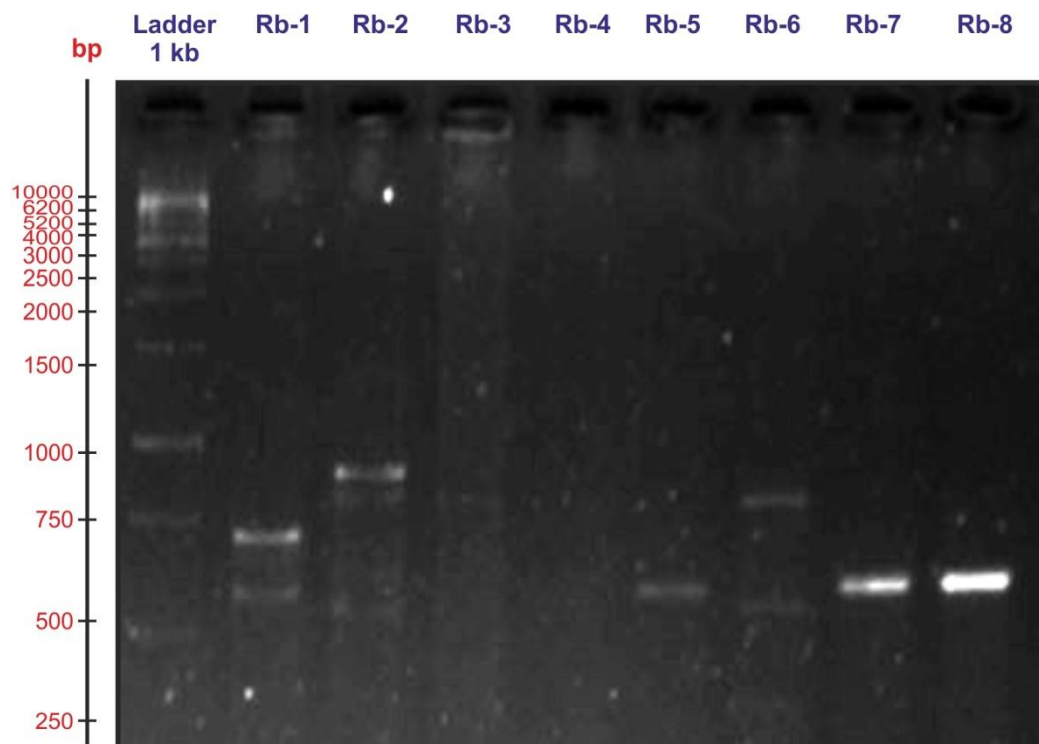


Plate 13. RAPD banding pattern of primer OPA-17

Band type 1 (890bp)	: This type of band was observed in isolate from Nagpur.
Band type 2 (800bp)	: This type of band was observed in isolates of Nagpur, Amravati and Sangli.
Band type 3 (700bp)	: This type of band observed in isolate of Akola.
Band type 4 (660bp)	: This type of band was observed in isolates from Kurundwad and Kini.
Band type 5 (620bp)	: This type of band was observed in isolate from Karanje.
Band type 6 (587bp)	: This type of band was observed in isolates from Akola and Sangli.
Band type 7 (550bp)	: This type of band was observed only in the isolate of Nagpur.

4.6 Binary similarity matrix

4.6.1 Binary similarity matrix for RAPD analysis

A binary similarity matrix of combined data from ten RAPD primers for the eight isolates *Rhizoctonia bataticola* isolates was prepared by scoring bands for presence or absence. DNA bands of same molecular weight were assumed to be identical.

On the basis of calculated similarity matrix the similarity between two genotypes can be predicted. The genotypes showing similarity index of “1” are presumed to be 100 per cent similar while that of “0” are 100 per cent genetically dissimilar. In present study the similarity coefficient value ranged from 0.7255 to 0.3492 across eight isolates of *Rhizoctonia bataticola* indicating high degree of variation in respect to genetic similarity. Genetic Similarity estimated (Jaccard's coefficient) based on RAPD banding pattern used for cluster analysis to present genetic relationship in the form of dendrogram (Fig.1). Jaccard's coefficient values for eight isolates of *Rhizoctonia bataticola* are presented in Table 19.

Table 19. Similarity coefficient for RAPD analysis

	Rb-1	Rb-2	Rb-3	Rb-4	Rb-5	Rb-6	Rb-7	Rb-8
Rb-1	1.0000							
Rb-2	0.4375	1.0000						
Rb-3	0.3684	0.4667	1.0000					
Rb-4	0.3492	0.3971	0.5882	1.0000				
Rb-5	0.3709	0.3571	0.5000	0.7255	1.0000			
Rb-6	0.3662	0.4459	0.4531	0.4286	0.5152	1.0000		
Rb-7	0.4754	0.4085	0.3650	0.4091	0.4531	0.4189	1.0000	
Rb-8	0.5000	0.5156	0.4310	0.4516	0.4754	0.4167	0.5079	1.0000

In the Dendrogram, Rb-4 (Umbraj) was found to have higher value of similarity coefficient (0.7255) whereas Rb-1 (Akola) was found to have lower value of similarity coefficient (0.3492). The isolates of *Rhizoctonia bataticola* were grouped into two major clusters. First group is named as cluster-A, includes isolates belonging to Akola, Kurundwad, Nagpur and Kini. Second group is named as cluster-B, includes Amravati, Umbraj, Karanje and Sangli.

It shows that Rb-4 (Umbraj) have higher value of similarity coefficient with Rb-5 (Karanje) whereas Rb-1 (Akola) have lower value of similarity coefficient with Rb-4 (Umbraj).

The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 72.55 to 34.92 per cent similarity coefficient.

Rajkumar *et al.* (2007) studied genetic diversity among 10 isolates of *Macrophomina phaseolina*, a causal agent of charcoal rot disease in sorghum using PCR-RAPD markers. From a set of 40 random primers tested, amplicon profiles of 15 were reproducible. A total of 149 amplicon levels, with an average of 9.9 bands per primer, were available for analysis of which 148 were polymorphic (99.3 %). UPGMA clustering of

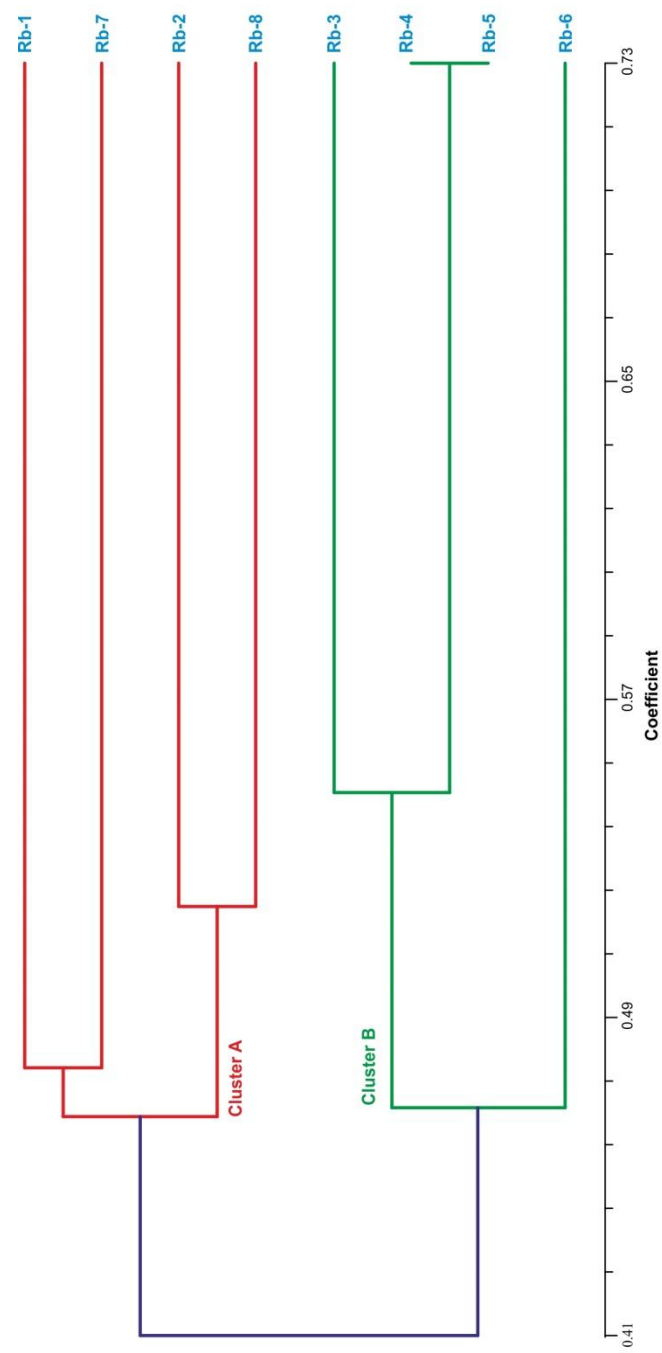


Fig. 1. The RAPD UPGMA dendrogram of eight isolates of *Rhizoctonia bataticola* based on Jaccard's similarity coefficient

data indicated the isolates shared varied levels of genetic similarity within a range of 0.14 to 0.72 similarities coefficient index and grouping of isolates was not related to sampling location

Monga *et al.* (2007) analyzed twenty five isolates of *Rhizoctonia bataticola* causal organism of cotton root rot using RAPD markers to study the relationship between molecular variability and pathogenicity. All the fifteen primers used generated scorable polymorphic bands for the isolates and classified the isolates into two broad groups. The group-1 consisted of isolate no. 16 having the genetic similarity 0.59 and found to be pathogenic (83.3 %). Group-2 was further clustered into five sub-groups. The largest sub-groups were II b and II c which consisted of 9 isolates each with 0.79-1.0 genetic similarity coefficient. The other sub-group IIa, II d and II e consisted of two isolates each having similarity coefficient ranged from 0.63-1.0. However, no strict correlation was observed between isolates grouped based on pathogenicity, morphological features and RAPD finger printing.

Pancheshwar *et al.* (2012) isolated the genomic DNA of 21 isolates of *Rhizoctonia bataticola* and subjected to amplification with RAPD markers for diversity analysis. Randomly selected 8 decamer primer amplified 64 RAPD marker loci. Out of these 64 bands, 29 bands (45.3 %) were polymorphic and remaining 35 bands (54.6 %) were monomorphic. The range of genetic diversity among *Rhizoctonia bataticola* isolates was 0.69 to 0.98. The cluster analysis showed 2 distinct groups among 21 isolates. They also suggested that any *Rhizoctonia bataticola* accessions under study did not made any group in respect to their geographical region.

Belkar and Gade (2016) isolated the genomic DNA of 20 isolates of *Rhizoctonia bataticola* and subjected to amplification with RAPD markers for diversity analysis. Randomly selected 20 decamer primer amplified 149 RAPD marker loci. Out of these 149 bands, 111 bands (74.49 %) were polymorphic. Similarity coefficient values for 20 accessions of *Rhizoctonia bataticola* in RAPDs marker were 0.36 - 0.99.

4.7 SSR primers selected for the study:

In all six primers were selected MB series from sigma technologies were used to evaluate the isolates of *Rhizoctonia bataticola* for SSR banding pattern. The Polymerase Chain Reaction (PCR) amplified products of each primer were resolved on 10 per cent poly acrylamide gel electrophoresis (PAGE) and the size of the amplified product was compared with DNA molecular weight marker 100 bp ladder.

Table 20. Per cent polymorphism observed in SSR primers

Sr. No.	Primer	Bands	Polymorphic bands	Monomorphic band	% Polymorphism
1	MB-2	12	9	3	75
2	MB-5	6	4	2	66.67
3	MB-9	4	4	0	100
4	MB-14	14	9	5	64.29
5	MB-17	8	7	1	87.50
6	MB-18	10	10	0	100
Total		54	43	11	79.62

4.8 SSR banding pattern

The polymorphic banding pattern was observed in six primers. The genetic variability in *Rhizoctonia bataticola* was observed in the SSR banding pattern and is given below.

The banding pattern observed in primer MB-2 is presented in Plate14. The primer amplified 12 amplicons among eight isolates of *Rhizoctonia bataticola*. The size of the amplicons amplified ranges from 778.46bp-190.38bp. The polymorphism observed in this primer was 75 %.The details of the 12 SSR band types are as follows.

Band type 1(778.46bp) : This type of band was observed in the isolates from Umbraj, Karanje, Kini.

Band type 2(675.39bp) : This type of band was observed in all isolates.

Band type 3(610.85bp)	: This type of band was observed in the isolates from Umbraj, Karanje and Kini.
Band type 4(490.87bp)	: This type of band was observed in all the isolates.
Band type 5(482.82bp)	: This type of band was observed only in the isolate of Kini.
Band type 6(419.91bp)	: This type of band was observed in all the isolates except Kini.
Band type 7(388.68bp)	: This type of band was observed only in the isolate from Kini.
Band type 8(320.74bp)	: This type of band was observed in the isolate from Akola and Kurundwad.
Band type 9(300.45bp)	: This type of band was observed in all the isolates.
Band type 10(286.51bp)	: This type of band was observed in the isolates from Akola, Nagpur, Kurundwad, Kini.
Band type 11(230.27bp)	: This type of band was observed in isolate of Umbraj.
Band type 12(190.38bp)	: This type of band was observed in the isolates from Nagpur, Karanje, Sangli, Kini.

The banding pattern observed in primer MB-5 is presented in Plate 14. The primer amplified 6 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 461.82bp to 151.12bp. The polymorphism observed in the primer was 66.67%. The details of the 6 SSR band types are as follows.

Band type 1(461.82bp)	: This type of band was observed in all isolates.
Band type 2(290.51bp)	: This type of band was observed in the isolates of Karanje, Sangli, Kini.
Band type 3(220.54bp)	: This type of band was observed in all the isolates.
Band type 4(205.84bp)	: This type of band was observed only in isolate of Akola.
Band type 5(152.21bp)	: This type of band was observed in isolate of Karanje and Sangli.
Band type 6(151.12bp)	: This type of band was observed in isolate of Akola, Karanje and Sangli.

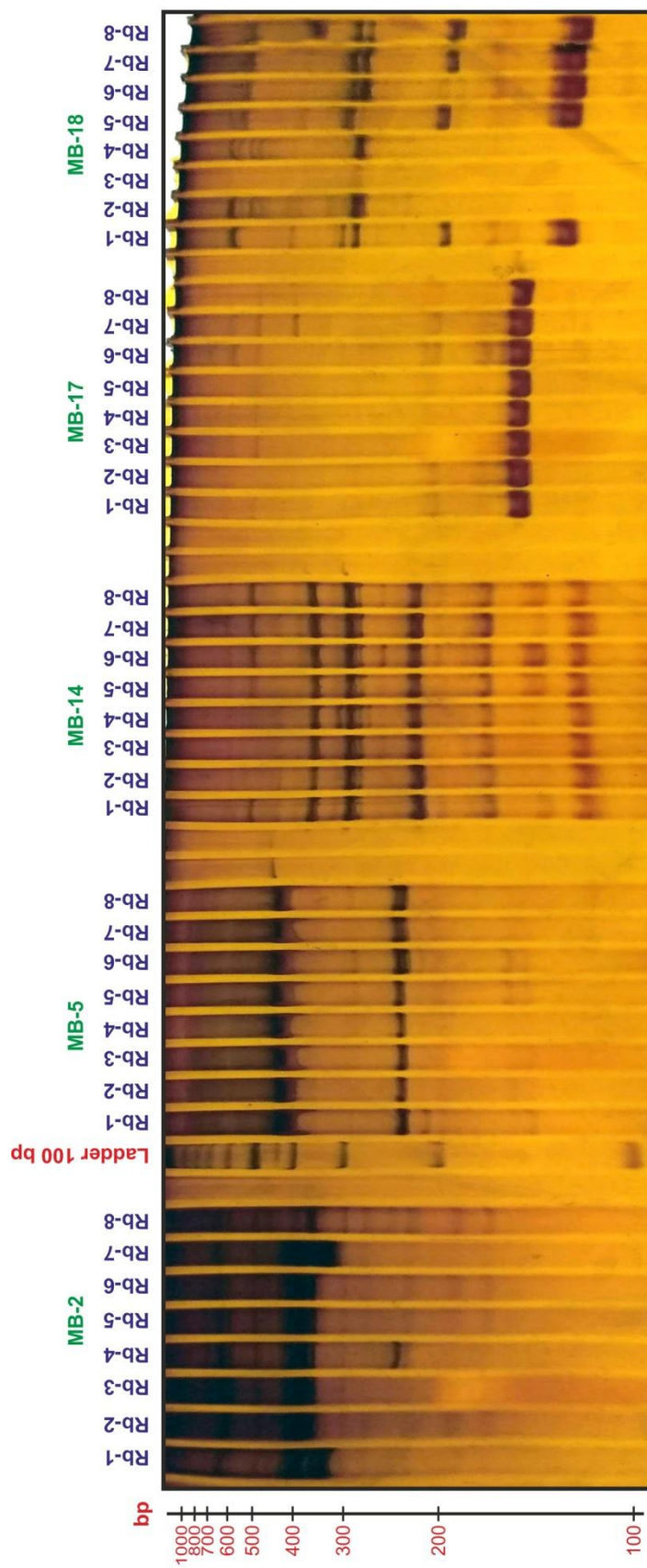


Plate 14. SSR banding pattern of primers MB-2, MB-5, MB-14, MB-17 and MB-18

The banding pattern observed in primer MB-9 is presented in Plate 15. The primer amplified 4 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 410.10bp to 100.16bp. The polymorphism observed in the primer was 100%. The details of the 4 SSR band types are as follows.

- Band type 1(410.10bp) : This type of band was observed in isolates of Amravati, Sangli, Kurundwad, Kini.
- Band type 2(204.17bp) : This type of band was observed in the isolates of Sangli and Kurundwad.
- Band type 3(128.18bp) : This type of band was observed in the isolates of Amravati, Umbraj, Sangli and Karanje.
- Band type 4(100.16bp) : This type of band was observed in all the isolates except Kini.

The banding pattern observed in primer MB-14 is presented in Plate 14. The primer amplified 14 amplicons among eight isolates of *Rhizoctonia bataticola*. The size of the amplicons amplified ranges from 575.82bp-130.40bp. The polymorphism observed in this primer was 64.29%. The details of the 14 SSR band types are as follows.

- Band type 1(575.82bp) : This type of band was observed in the isolates from Akola, Karanje, Sangli, Kurundwad, Kini.
- Band type 2(486.30bp) : This type of band was observed in isolates of Akola, Nagpur, Kurundwad.
- Band type 3(410.38bp) : This type of band was observed only in the isolate from Kurundwad.
- Band type 4(390.88bp) : This type of band was observed in all the isolates.
- Band type 5(320.74bp) : This type of band was observed in the isolate from Umbraj.
- Band type 6(310.45bp) : This type of band was observed in all the isolates.
- Band type 7(289.51bp) : This type of band was observed in only the isolate of Umbraj.

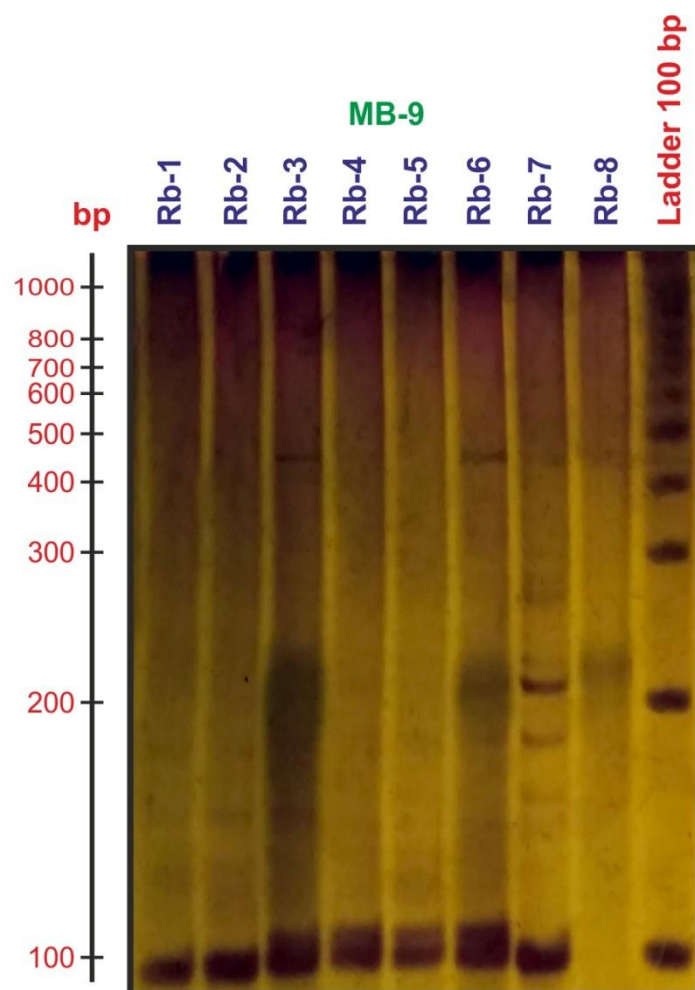


Plate 15. SSR banding pattern of primer MB-9

Band type 8(260.09bp)	: This type of band was observed in the isolates from Sangli, Kini.
Band type 9(210.15bp)	: This type of band was observed in all the isolates.
Band type 10(205.70bp)	: This type of band was observed in the isolates of Akola, Sangli, Kurundwad and Kini.
Band type 11(160.57bp)	: This type of band was observed in all the isolates.
Band type 12(153.39bp)	: This type of band was observed in the isolates of Karanje, Sangli and Kini.
Band type 13(150.56bp)	: This type of band was observed in the isolates of Akola, Karanje, Sangli, Kurundwad, Kini.
Band type 14(130.40bp)	: This type of band was observed in all the isolates.

The banding pattern observed in primer MB-17 is presented in Plate 14. The primer amplified 8 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 810.04bp to 150.07bp. The polymorphism observed in the primer was 87.50%. The details of the 8 SSR band types are as follows.

Band type 1 (810.04bp)	: This type of band was observed only in the isolate of Sangli.
Band type 2 (788.82bp)	: This type of band was observed in the isolate from Sangli and Kurundwad.
Band type 3 (553.97bp)	: This type of band was observed in isolates from Akola, Amravati, Umbraj, Sangli, Kurundwad.
Band type 4 (430.86bp)	: This type of band was observed only in isolate from Kurundwad.
Band type 5 (208.71bp)	: This type of band was observed only in isolate from Sangli.
Band type 6 (205.84bp)	: This type of band was observed in isolates from Akola, Nagpur, Sangli and Kurundwad.
Band type 7 (160.70bp)	: This type of band was observed only in the isolate of Sangli.
Band type 8 (150.07bp)	: This type of band was observed in all the isolates.

The banding pattern observed in primer MB-18 is presented in Plate 14. The primer amplified 10 amplicons from eight isolates of *Rhizoctonia bataticola*. The size of the amplicons ranged from 778.46bp to 140.79bp. The polymorphism observed in the primer was 100%. The details of the 10 SSR band types are as follows.

- Band type 1(778.46bp) : This type of band was observed in isolates of Umbraj and Karanje.
- Band type 2(700.75bp) : This type of band was observed in all the isolates except Amravati.
- Band type 3(620.85bp) : This type of band was observed in the isolates of Nagpur and Umbraj.
- Band type 4(490.82bp) : This type of band was observed in isolates of Akola, Nagpur, Karanje, Kurundwad and Kini.
- Band type 5(380.88bp) : This type of band was observed only in isolate of Nagpur.
- Band type 6(340.74bp) : This type of band was observed in isolates of Akola, Karanje, Sangli, Kurundwad, Kini.
- Band type 7(320.68bp) : This type of band was observed in isolates of Kurundwad, Akola, Nagpur, Umbraj, Sangli, Kini.
- Band type 8(205.63bp) : This type of band was observed in isolates of Akola, Karanje, Kurundwad and Kini.
- Band type 9(160.37bp) : This type of band was observed in isolates of Akola, Karanje, Sangli, Kurundwad and Kini.
- Band type 10(140.79bp) : This type of band was observed in isolates of Akola, Karanje, Sangli, Kurundwad and Kini.

4.9 Binary similarity matrix

4.9.1 Binary similarity matrix for SSR analysis

A binary similarity matrix of combined data from six SSR primers for eight isolates of *Rhizoctonia bataticola* isolates was prepared by scoring bands for presence or absence. DNA bands of same molecular weight were assumed to be identical.

On the basis of calculated matrix the similarity between two genotypes can be predicted. The genotypes showing similarity index of “1” are presumed to be 100 per cent similar while that of “0” are 100 per cent genetically dissimilar. In present study the similarity coefficient value ranged from 0.3529 to 0.7647 across eight isolates of *Rhizoctonia bataticola* indicating high degree of variation in respect to genetic similarity. Genetic similarity estimated (Jaccard's coefficient) based on SSR banding pattern used for clusters analysis to present genetic relationship in the form of dendrogram (Fig. 2) Jaccard's coefficient values for eight isolates of *Rhizoctonia bataticola* are presented in Table 21.

Table 21. Similarity coefficient for SSR analysis

	Rb-1	Rb-2	Rb-3	Rb-4	Rb-5	Rb-6	Rb-7	Rb-8
Rb-1	1.0000							
Rb-2	0.6129	1.0000						
Rb-3	0.4666	0.5200	1.0000					
Rb-4	0.4444	0.5333	0.6000	1.0000				
Rb-5	0.5833	0.4571	0.4516	0.5142	1.0000			
Rb-6	0.5365	0.4250	0.4571	0.4047	0.6000	1.0000		
Rb-7	0.7647	0.5428	0.4545	0.4000	0.4878	0.5952	1.0000	
Rb-8	0.5263	0.4444	0.3529	0.3846	0.6388	0.5476	0.5500	1.0000

In the Dendrogram, Rb-1 (Akola) was found to have higher value of similarity coefficient (0.7647) whereas Rb-3 (Amravati) was found to have lower value of similarity coefficient (0.3529). The isolates of *Rhizoctonia bataticola* were grouped into two major clusters. First group is named as cluster-A, includes isolates belonging to Akola, Kurundwad, Nagpur, Karanje, Kini and Sangli. Second group is named as cluster-B, includes Amravati and Umbraj.

It shows that Rb-1 (Akola) have higher value of similarity coefficient with Rb-7 (Kurundwad) whereas Rb-3 (Amravati) have lower value of similarity coefficient with Rb-8 (Kini).

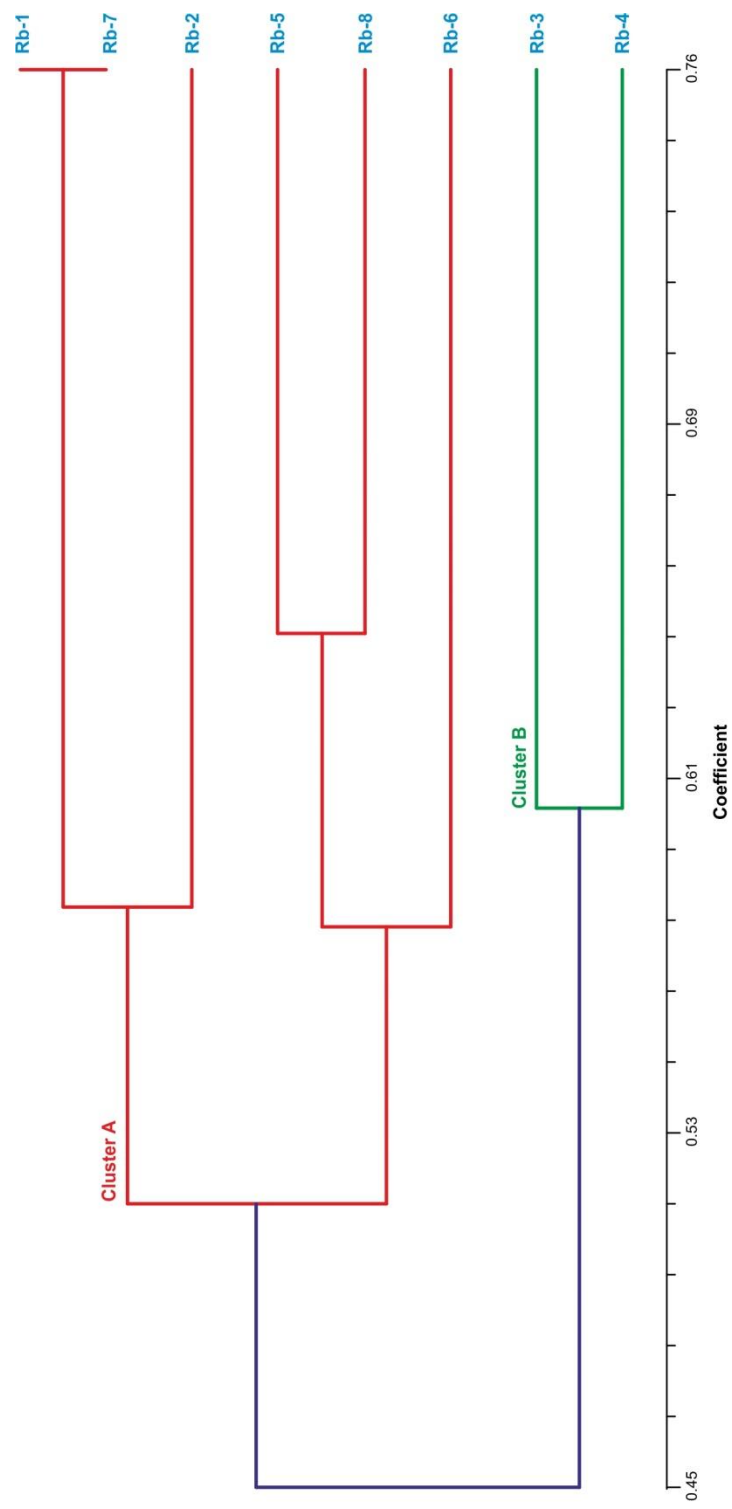


Fig. 2. The SSR UPGMA dendrogram of eight isolates of *Rhizoctonia bataticola* based on Jaccard's similarity coefficient

The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 35.29-76.47 per cent similarity coefficient.

Bogale *et al.* (2005) describe nine simple sequence repeat (SSR) markers developed for studying *Fusarium oxysporum*. Allelic diversity at the nine loci ranged from 0.003 to 0.895, with a total of 71 alleles among 64 isolates. These markers will facilitate studies on relationships amongst isolates of *Fusarium oxysporum*.

Jana *et al.* (2005) used SSR markers for the first time for the characterization of genetic variability of different populations of *Macrophomina phaseolina* obtained from soybean and cotton grown in India and USA. Three groups were obtained with UPGMA among all the isolates except three isolates. The major group (group I) consisted of 27 isolates, having >80% similarity and all were obtained from soybean plant roots. A second group (group II) of 3 isolates also obtained from soybean roots was formed at 52% similarity level. Similarly, isolates obtained from cotton roots (collected in the USA) formed a distinct group (group III) above 97% genetic similarity level. Thus all isolates collected from soybean and cotton roots were clustered separately in distinct groups using UPGMA.

Bimal *et al.* (2013) investigated that 20 ISSR primers amplified 174 markers. Out of them 122 loci were polymorphic 70.11%, while 52 loci were monomorphic 29.89%. A dendrogram was generated by UPGMA clusters analysis based on Jaccard's similarity coefficient. Estimated genetic similarity among 18 soybean varieties using 20 ISSR primers, were 0.71-0.95.

CHAPTER V

SUMMARY AND CONCLUSIONS

Chickpea (*Cicer arietinum*) is the world's third most important pulse crop. The major constraints limiting chickpea production is difficulty in managing diseases caused by soil borne pathogens viz., dry root rot (*Rhizoctonia bataticola*), Fusarium wilt (*Fusarium oxysporum* f.sp. *ciceri*). The losses in chickpea due to *Rhizoctonia bataticola* may be as high as 60 per cent.

The investigation was carried out in the laboratory of the Department of Plant Pathology and Biotechnology Centre, Dr. PDKV, Akola. The eight isolates of *Rhizoctonia bataticola* were collected from different districts of Maharashtra viz., Akola, Nagpur, Amravati, Satara (Karanje), Satara (Umbraj), Sangli, Sangli (Kurundwad) and Kolhapur (Kini).

5.1 Pathogenicity

The eight isolates of chickpea dry root rot pathogen was collected from different districts of Maharashtra viz., Akola, Nagpur, Amravati, Satara (Karanje), Satara (Umbraj), Sangli, Sangli (Kurundwad) and Kolhapur (Kini). Pathogenicity was tested on JG-62 of which two isolates were strongly pathogenic, six were moderately pathogenic.

5.2 Random Amplified Polymorphic DNA (RAPD)

The genetic diversity was studied amongst the eight isolates of *Rhizoctonia bataticola* using fifteen RAPD primers. Among them 10 primers produced 101 scorable bands. Out of 101 bands, 85 bands were polymorphic and average level of polymorphism was 84.16 %. The primer OPB-14 amplified maximum 14 bands within the size 1951.44bp to 336bp, while primer OPB-11 and OPF-4 amplified minimum of 6 bands within the

size 2159.78bp to 623bp and 950bp to 262.17bp, respectively. The ten RAPD primers showed 16 monomorphic bands.

In the Dendrogram, Rb-4 (Umbraj) was found to have higher value of similarity coefficient (0.7255) whereas Rb-1 (Akola) was found to have lower value of similarity coefficient (0.3492). The isolates of *Rhizoctonia bataticola* were grouped into two major clusters. First group is named as cluster-A, includes isolates belonging to Akola, Kurundwad, Nagpur and Kini. Second group is named as cluster-B, includes Amravati, Umbraj, Karanje and Sangli.

It shows that Rb-4 (Umbraj) have higher value of similarity coefficient with Rb-5 (Karanje) whereas Rb-1 (Akola) have lower value of similarity coefficient with Rb-4 (Umbraj).

The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 34.92 to 72.55 per cent similarity coefficient.

5.3 Simple Sequence Repeat (SSR)

Genetic diversity was studied among the eight isolates using 6 SSR primers of MB series. Six primers produced 54 bands. Out of 54 bands, 43 bands were polymorphic and average level of polymorphism was 79.62 %. The primer MB-14 amplified maximum of 14 bands within the size 130.40bp to 575.82bp, while primer MB-9 amplified minimum of 4 bands within the size 100.16bp to 410.10bp, respectively. The 4 SSR primers MB-2 (300.45bp, 490.87bp, 675.39bp), MB-5 (220.54bp, 461.82bp), MB-14 (130.40bp, 160.57bp, 210.15bp, 310.45bp, 390.88bp), MB-17 (150.07bp) showed 11 monomorphic bands. Primers MB-9 and MB-18 showed 100 % polymorphic bands.

In the Dendrogram, Rb-1 (Akola) was found to have higher value of similarity coefficient (0.7647) whereas Rb-3 (Amravati) was found to have lower value of similarity coefficient (0.3529). The isolates of *Rhizoctonia bataticola* were grouped into two major clusters. First group is

named as cluster-A, includes isolates belonging to Akola, Kurundwad, Nagpur, Kini, Karanje and Sangli. Second group is named as cluster-B, includes Amravati and Umbraj.

It shows that Rb-1 (Akola) have higher value of similarity coefficient with Rb-7 (Kurundwad) whereas Rb-3 (Amravati) have lower value of similarity coefficient with Rb-8 (Kini).

The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 35.29-76.47 per cent similarity coefficient.

Conclusions

The following conclusions can be drawn from the present investigation

1. In pathogenecity test that all the 8 isolates of *Rhizoctonia bataticola* proved to be pathogenic to cultivar JG-62, of which, 2 isolates were strongly pathogenic (51-70%) and 6 isolates were moderately pathogenic (21-50%).
2. Molecular identification by ITS marker is good tool in detection of *Rhizoctonia bataticola*. All the isolates of *Rhizoctonia bataticola* yielded the 500 bp band with the ITS marker, hence confirm the identity of *Rhizoctonia bataticola* were as per the result of Aghakhani and Dubey (2009^b).
3. The RAPD analysis total 15 primers were screened against eight isolates of *Rhizoctonia bataticola*. Out of 15 primers 10 primers amplified scorable banding pattern. Total 101 amplicons amplified with 10 primers out of which 85 amplicons were polymorphic. Level of polymorphism for RAPD analysis was 84.16%.
4. The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 34.92 to 72.55 per cent similarity coefficient for RAPD primers. The RAPD analysis shows high molecular variability among the isolates and grouped into two major clusters A and B.

5. The SSR analysis total 6 primers were screened against eight isolates of *Rhizoctonia bataticola*. Total 54 amplicons amplified with 6 primers out of which 43 amplicons were polymorphic. Level of polymorphism for SSR analysis was 79.62%.
6. The similarity matrix indicated that eight isolates of *Rhizoctonia bataticola* exhibited in between 35.29-76.47 per cent similarity coefficient for SSR primers.

CHAPTER VI

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5. Academic Qualification :

Sl. No.	Name of Degrees awarded	Year in which obtained	Division/ Class	Name of awarding University	Subjects
1.	B.Sc. (Agri.)	2015	First Class	MPKV, Rahuri	Agriculture and allied subject

6. Research papers published (if any) : NIL
7. Field of Interest (in which you desire to work) : Research and Development in Plant Pathology and Agril. Biotechnology.

Place : Akola

Date : / /2017

(Gadekar Archana Arun)
Signature of Student

APPENDIX I

List of bands amplified against RAPD primers

Primers	bp	Rb-1	Rb-2	Rb-3	Rb-4	Rb-5	Rb-6	Rb-7	Rb-8
OPA-2	4860.00bp	0	1	0	0	0	0	0	1
	2302.58bp	1	1	1	1	1	1	1	1
	2013.938bp	1	1	0	0	1	0	1	1
	1784.20bp	0	0	0	0	0	0	1	0
	1442.88bp	0	1	1	1	1	1	0	1
	1202.93bp	1	1	1	0	1	1	1	1
	954.11bp	1	1	0	0	0	1	1	1
	834.52bp	0	1	0	0	0	0	0	1
	740.52bp	1	1	0	0	0	1	1	1
	628.44bp	1	1	1	1	1	1	1	1
	569.88bp	1	1	0	0	0	0	0	0
	544.13bp	1	1	1	1	0	1	1	1
	381.82bp	1	1	0	0	0	1	1	1
OPA-5	2714.82bp	0	0	0	0	0	0	1	0
	1847.37bp	0	1	0	0	0	1	1	0
	1624.46bp	0	1	0	0	0	0	0	0
	1409.13bp	0	0	0	1	1	1	0	0
	1332.54bp	0	1	0	0	0	1	1	0
	1210.00bp	0	1	1	1	0	0	0	0
	900.00bp	0	1	0	0	0	1	1	1
	860.07bp	0	1	1	1	1	1	1	1
	691.59bp	0	0	0	1	1	1	0	1
OPA-7	4860.41bp	1	1	0	0	0	0	1	1
	3482.15bp	1	0	0	1	1	0	1	0
	2176.09bp	1	1	1	1	1	1	1	1
	1913.98bp	1	1	1	0	0	0	0	1
	1697.40bp	1	0	0	1	1	1	1	0
	1402.93bp	1	1	1	1	1	1	1	1
	1108.19bp	1	1	0	0	0	0	0	1
	1026.00bp	0	1	0	0	0	0	1	0
	890.74bp	0	0	0	1	1	1	1	0
	644.79bp	0	1	0	0	0	0	0	1
	628.44bp	1	1	1	1	1	1	1	1
	500.88bp	0	0	1	0	0	0	1	0
OPA-9	2500bp	0	0	1	1	1	1	0	1
	2239bp	0	0	0	0	0	1	0	1
	1655bp	0	1	1	1	1	1	1	0
	1477.14bp	0	0	0	0	0	1	1	1
	1279.93bp	0	0	0	0	0	0	1	0
	1171.30bp	0	1	1	1	1	1	0	1
	1034.75bp	0	0	0	0	0	1	0	0
	775.84bp	0	0	0	0	0	0	1	0
	684.09bp	1	1	1	1	1	1	1	1
	694.93bp	1	0	0	0	0	0	0	0
OPA-10	2541.5bp	0	1	1	1	1	1	0	0
	2329.62bp	0	1	0	1	1	0	0	0
	1577.66bp	1	1	1	1	1	1	1	1
	1271.30bp	0	0	0	0	1	1	0	0
	858.85bp	0	0	0	0	0	0	0	1
	664.09bp	1	1	1	1	1	1	1	1

Primers	bp	Rb-1	Rb-2	Rb-3	Rb-4	Rb-5	Rb-6	Rb-7	Rb-8
	626.68bp	1	0	0	0	0	0	0	0
	519.81bp	0	0	0	0	1	1	0	0
	507.02bp	0	0	0	0	1	1	0	0
	460.73bp	0	0	0	0	1	1	1	0
	412.04bp	0	0	0	1	1	1	0	0
OPA-13	2627.234bp	0	0	1	1	1	0	1	0
	2055.99bp	0	0	0	0	0	0	1	1
	1784.40bp	1	1	1	1	1	1	1	1
	1678.67bp	0	1	1	1	0	0	0	0
	1521.83bp	0	1	1	1	0	0	0	0
	1391.87bp	1	0	0	0	0	0	0	0
	1334.43bp	1	1	1	1	1	1	1	1
	1148.44bp	1	0	0	0	0	0	0	1
	909.51bp	1	0	0	0	0	0	1	0
	852.00bp	1	1	0	0	0	1	0	0
	810.00bp	0	0	0	0	0	1	0	0
	770.00bp	1	1	1	1	1	1	0	1
	632.00bp	1	1	1	1	1	1	1	1
OPB-11	2159.78bp	0	0	1	1	1	1	0	0
	1402.27bp	0	0	0	1	1	0	1	1
	1200bp	0	0	0	1	0	0	0	0
	1000bp	0	0	0	1	0	0	0	0
	782.00bp	0	0	0	1	0	0	0	0
	623.00bp	0	0	0	0	0	0	1	0
OPB-14	1951.44bp	0	1	0	0	0	0	0	0
	1565.92bp	0	0	0	0	0	1	0	0
	1331.87bp	0	1	0	0	0	1	0	0
	1100.00bp	1	1	1	1	1	1	1	1
	950.00bp	1	1	1	1	1	1	1	1
	827.33bp	0	0	1	0	0	1	0	0
	800.00bp	0	0	0	0	0	1	0	0
	790.00bp	0	0	0	0	0	1	0	0
	760.00bp	0	1	0	0	0	0	0	0
	657.00bp	1	0	1	0	0	1	1	0
	590.00bp	1	1	1	1	1	1	1	1
	487.00bp	1	0	1	1	1	0	1	1
	400.00bp	0	0	1	0	0	1	0	0
	336.00bp	0	0	0	0	0	1	0	0
OPF-4	950.00bp	0	0	0	1	1	0	1	1
	888.84bp	1	0	0	1	1	0	1	1
	589.41bp	0	0	1	1	1	0	0	1
	432.89bp	1	0	0	0	0	0	0	0
	349.17bp	1	1	0	1	1	1	1	1
	262.17bp	1	1	1	1	1	1	1	1
OPA-17	890.00bp	0	1	0	0	0	0	0	0
	800.00bp	0	1	1	0	0	1	0	0
	700.00bp	1	0	0	0	0	0	0	0
	660.00bp	0	0	0	0	0	0	1	1
	620.00bp	0	0	0	0	1	0	0	0
	587.00bp	1	0	0	0	0	1	0	0
	550.00bp	0	1	0	0	0	0	0	0

APPENDIX II

List of bands amplified against SSR primers

Primers	bp	Rb-1	Rb-2	Rb-3	Rb-4	Rb-5	Rb-6	Rb-7	Rb-8
MB-2	778.46bp	0	0	0	1	1	0	0	1
	675.39bp	1	1	1	1	1	1	1	1
	610.85bp	0	0	0	1	1	0	0	1
	490.87bp	1	1	1	1	1	1	1	1
	482.82bp	0	0	0	0	0	0	0	1
	419.91bp	1	1	1	1	1	1	1	0
	388.68bp	0	0	0	0	0	0	0	1
	320.74bp	1	0	0	0	0	0	1	0
	300.45bp	1	1	1	1	1	1	1	1
	286.51bp	1	1	0	0	0	0	1	1
MB-5	230.27bp	0	0	0	1	0	0	0	0
	190.38bp	0	1	0	0	1	1	0	1
	461.82bp	1	1	1	1	1	1	1	1
	290.51bp	0	0	0	0	1	1	0	1
	220.54bp	1	1	1	1	1	1	1	1
	205.84bp	1	0	0	0	0	0	0	0
	152.21bp	0	0	0	0	1	1	0	0
	151.12bp	1	0	0	0	1	1	0	0
	410.10bp	0	0	1	0	0	1	1	1
	204.17bp	0	0	0	0	0	1	1	0
MB-9	128.18bp	0	0	1	1	1	1	0	0
	100.16bp	1	1	1	1	1	1	1	0
	575.82bp	1	0	0	0	1	1	1	1
	486.30bp	1	1	0	0	0	0	1	0
	410.38bp	0	0	0	0	0	0	1	0
	390.88bp	1	1	1	1	1	1	1	1
	320.74bp	0	0	0	1	0	0	0	0
	310.45bp	1	1	1	1	1	1	1	1
	289.51bp	0	0	0	1	0	0	0	0
	260.09bp	0	0	0	0	0	1	0	1
MB-14	210.15bp	1	1	1	1	1	1	1	1
	205.70bp	0	0	0	0	0	1	1	1
	160.57bp	1	1	1	1	1	1	1	1
	153.39bp	0	0	0	0	1	1	0	1
	150.56bp	1	0	0	0	1	1	1	1
	130.40bp	1	1	1	1	1	1	1	1
	810.04bp	0	0	0	0	0	1	0	0
	788.82bp	0	0	0	0	0	1	1	0
	553.97bp	1	0	1	1	0	1	1	0
	430.86bp	0	0	0	0	0	0	1	0
MB-17	208.71bp	0	0	0	0	0	1	0	0
	205.84bp	1	1	0	0	0	1	1	0
	160.70bp	0	0	0	0	0	1	0	0
	150.07bp	1	1	1	1	1	1	1	1
	778.46bp	0	0	0	1	1	0	0	0
	700.75bp	1	1	0	1	1	1	1	1
	620.85bp	0	1	0	1	0	0	0	0
	490.82bp	1	1	0	0	1	0	1	1
	380.88bp	0	1	0	0	0	0	0	0
	340.74bp	1	0	0	0	1	1	1	1
MB-18	320.68bp	1	1	0	1	0	1	1	1
	205.63bp	1	0	0	0	1	0	1	1
	160.37bp	1	0	0	0	1	1	1	1
	140.79bp	1	0	0	0	1	1	1	1